

# TESTABLE IMPLICATIONS OF PARAMETRIC ASSUMPTIONS IN ORDERED RANDOM UTILITY MODELS

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**ABSTRACT.** We establish the testable implications of ordered random utility models under nonparametric, semi-nonparametric, and parametric assumptions. After characterizing these models in a continuous setting and in the absence of fundamental parametric restrictions, we show that assumptions about the type distribution alone are immaterial, while assumptions about the map linking types to utilities are relevant only insofar as they restrict the class of utilities at stake. Importantly, the joint presence of such parametric assumptions, as is common practice, further restricts the empirical content of the model. We then provide a characterization of parametric ordered-logit models. Finally, we apply our results to economically relevant settings.

**Keywords:** Ordered random utility model; empirical content; parametric restrictions; ordered logit; cumulative logit; cumulative log-odds.

**JEL classification numbers:** C00; D00.

## 1. INTRODUCTION

In many settings, decision problems are ordered and variation in choices can be intuitively explained as the result of variation in an underlying ordered latent variable. This simple structure is a fundamental instrument for empirical research, spanning diverse economic areas such as health, finance, labor, welfare, management, insurance, political economy, networks, and gender.<sup>1</sup> To formalize this notion, consider a latent variable on the real line, and let us refer to each of its values as a type. In what we refer to as Ordered Random Utility Models (ORUMs), choice data is the result of

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<sup>1</sup>See, respectively, Barsky, Juster, Kimball, and Shapiro (1997), Kaplan and Zingales (1997), Blau and Hagy (1998), Campante and Yanagizawa-Drott (2015), Cummings (2004), Cohen and Einav (2007), Besley and Persson (2011), Bailey, Cao, Kuchler, and Stroebel (2018), and Carlana (2019).

the interaction of the following two components: (i) a type-utility map where higher types are associated with utility functions generating higher choices, and (ii) a type distribution describing the prevalence of each type.

The empirical use of ORUMs often involves a number of parametric assumptions on one or both of the components of the model. To illustrate, regarding the first component, applications to political or insurance choices are often built by referring to type  $t$  as the Euclidean utility centered at  $t$  or the expected utility with a relative risk aversion coefficient equal to  $t$ , respectively.<sup>2</sup> Regarding the second component, the most common assumptions may involve logistic (or Gaussian) variation in the latent variable, leading to the so-called ordered-logit (or ordered-probit) models. And in many cases, both components are jointly restricted, leading to a fully parametric model. It is then critical to carefully examine when and how parametric specifications bear down on the space of datasets that can be explained by ORUMs and, in each case, what type of properties characterize the corresponding model. This is the purpose of this paper.

We present our results within the context of cumulative choice data over an arbitrary collection of continuous decision problems. We start, in Section 3, with the nonparametric case and show that the model can be understood by means of a standard, deterministic notion of rationalizability. In a nutshell, suppose that for each probability value  $p \in (0, 1)$ , we construct from data the quantile choice function  $c^p$ , i.e.,  $c_j^p$  represents the alternative that first attains cumulative choice above  $p$  in decision problem  $A_j$ . Theorem 1 shows that data can be explained by an ORUM if and only if every quantile choice function is acyclical. The proof is rather direct but, fundamentally for our purposes, it is instrumental for understanding the exact content of semi-nonparametric ORUMs and flexible enough to be used in a variety of applied settings.

Section 4 is devoted to the semi-nonparametric analysis. Building upon the proof of Theorem 1, we show in Corollary 1 that constraining only the type distribution comes at no cost; as long as the type-utility map is not fixed, the analyst can freely reduce the dimensionality of the statistical distribution describing the latent variable. We then consider the empirical content of semi-nonparametric models that constrain the type-utility map but not the type distribution. Naturally, this restriction is relevant because we are now forced to use only the sub-class of utilities specified by the parametrization, and this limits our ability to explain patterns of choice across menus. However, we show in Theorem 2 that, modulo the sub-class of utilities in the parametrization, the assumption brings no further cost. Formally, if the type distribution remains unrestricted, fixing the type-utility map merely requires properly adjusting the deterministic rationalizability property used in Theorem 1 for it to apply to the corresponding sub-class of utilities in the parametrization.

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<sup>2</sup>This type of parametric restriction is often referred to as semi-nonparametric (see, e.g., Barseghyan, Molinari, and Thirkettle (2021)). To simplify our exposition, we use the same term when the dimensionality of one, but not the other, component is constrained.

Section 5 brings the fully parametric analysis. In essence, fixing both the type-utility map and the type distribution fundamentally restricts the model, and does so in ways that cannot be readily described, even with the adjustment in Theorem 2. As a result, we are forced to analyze the properties of parametric ORUMs independently. Given the preponderance of ordered-logit models in the empirical literature, we study ORUMs in which, having fixed a type-utility map, the type distribution is assumed to be logistic.<sup>3</sup> We show in Theorem 3 that two simple properties, which we call corner extremeness (CE) and cumulative log-odds additivity (CLA), characterize these models. CE imposes that a corner alternative can receive a non-null choice probability if and only if this alternative is not dominated, i.e., considered sub-optimal by all utilities in the class. CLA uses the well-known notion of cumulative log-odds, i.e., the log-ratio of masses below and above a given alternative. The property states that equal sums of types must lead to equal sums of cumulative log-odds. Theorem 3 establishes the first characterization result of ordered-logit models, giving foundations to a popular tool in the empirical literature.

To illustrate our results from a more applied point of view, we study in detail two distinct applications to choices over arbitrary collections of linear budget sets: the first involving lotteries, and the second involving inter-personal allocations. Section 6.1 elaborates on the application to lotteries, with Corollary 2 providing nonparametric (as in Theorem 1) and semi-nonparametric results (as in Theorem 2) for ORUMs using expected utilities and CRRA expected utilities, respectively. Building upon Theorem 3, Corollary 3 characterizes the ordered-logit model that arises from logistic variation of the CRRA coefficient. Section 6.2 reproduces this analysis for the case of inter-personal allocations, where the semi-nonparametric and parametric cases adopt the CES functional form in modeling variation in altruism. In Appendix D we provide an empirical illustration of these results using simulated datasets that are representative of influential empirical papers in the literature.

## 2. RELATED LITERATURE

This paper contributes to the study of stochastic choice models in general, and to the literature on random utility models (RUMs) in particular. Classic works include Luce (1959), Block and Marschak (1960), and McFadden and Richter (1990). Here, we study RUMs with an ordered structure; see Small (1987) for an early study, Train (2009) for an econometric treatment, and Greene and Hensher (2010) for the applicability of this type of model to a variety of economic questions. RUMs with an ordered structure have recently been studied from various theoretical perspectives in Apesteguia, Ballester, and Lu (2017), Barseghyan, Molinari, and Thirkettle (2021), Tserenjigmid (2021), Turansick (2022), Valkanova (2022), Apesteguia and Ballester (2023), Filiz-Ozbay and Masatlioglu (2023), Masatlioglu and Vu (2023), Petri (2023, 2024), Caliori and Petri

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<sup>3</sup>The majority of works cited in footnote 1 adopt the ordered-logit format, which gives a good sense of the popularity of the model.

(2024), Yildiz (2024), Apesteguia, Ballester and Gutierrez-Daza (2025), and Chambers, Masatlioglu, and Yildiz (2025).

Part of this paper is devoted to establishing choice-based foundations for ORUMs with varying degrees of parametric assumptions.<sup>4</sup> Apesteguia, Ballester, and Lu (2017) provide the first axiomatization of a RUM with an ordered structure, which they call SCRUM. Their analysis is discrete and nonparametric, with unrestricted variation over single-crossing families of utilities. Their main result can be seen as a specific case of Theorem 1 because, for their analysis to hold, they need to observe data on every subset of a linearly ordered set  $X$ . Also within a discrete setting, Apesteguia and Ballester (2023) relax data requirements by working, as in the current paper, with arbitrary domains of ordered menus. Their main results, however, operate exclusively in the semi-nonparametric realm, in which the type-utility map is fixed. The present paper contributes to this literature in several ways. First, we establish axiomatic foundations for nonparametric, semi-nonparametric, and parametric versions of ORUMs, all with arbitrary data. Second, by adopting a common setting across all parametric versions, we are able to establish the differential empirical implications of parametric assumptions. Third, we provide, for the first time, an axiomatic treatment of the often-used parametric ordered logit, an analysis that requires the development of novel techniques drawn from the statistical literature. Finally, we adopt the empirically relevant but severely understudied continuous setting.<sup>5</sup>

Another relevant strand of literature pertains to the nonparametric identification of models that are non-additive in the error term, and the standard practice of normalization. To illustrate the connection to our results, consider Matzkin (2003), where a model of the form  $Y = m(X, \alpha)$ , with observables  $X$  and  $Y$  and a non-additive unobserved error term  $\alpha$ , is studied. The function  $m$  is assumed to be monotone in the latent variable  $\alpha$ . Our framework follows this approach by considering a function  $m$  that selects the element  $Y$  optimizing the utility function of type  $\alpha$ , given the menu characteristics  $X$ .<sup>6</sup> As per the usual normalization analysis, Lemma 1 in Matzkin (2003) addresses the joint identification of the function  $m$  and the distribution of the error term  $\alpha$ ; essentially, monotonic transformations of the pair  $(m, \alpha)$  yield the same

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<sup>4</sup>The literature providing choice-based foundations for other stochastic models is extensive. Some recent contributions include Gul and Pesendorfer (2006), Manzini and Mariotti (2014), Caplin and Dean (2015), Fudenberg, Iijima, and Strzalecki (2015), Matejka and McKay (2015), Brady and Rehbeck (2016), Ahn, Echenique, and Saito (2018), Lu and Saito (2018), Cerreia-Vioglio, Dillenberger, Ortoleva, and Riella (2019), Frick, Iijima, and Strzalecki (2019), Natenzon (2019), Cattaneo, Ma, Masatlioglu, and Suleymanov (2020), Alós-Ferrer, Fehr, and Netzer (2021), Kovach and Tserenjigmid (2022), Cerreia-Vioglio, Maccheroni, Marinacci, and Rustichini (2023), He and Natenzon (2023), or Alós-Ferrer and Mihm (2025).

<sup>5</sup>Our results can also be derived in the discrete choice setting. Appendix B does so and discusses in more detail the connection with the mentioned papers.

<sup>6</sup>One of the examples discussed in Matzkin (2003) is that of random demand over linear budget sets, which is closely related to the settings in our applications.

data. Despite the fundamental differences between this literature and our results, the logic of normalization is the key to extending our Theorem 1 to our Corollary 1; we elaborate later in the paper on how Corollary 1 can be viewed as a combination of Theorem 1 and normalization. Since the primary focus of the current paper is characterization rather than identification, we contribute to this literature in several ways. Most notably, we provide a precise description of the empirical content of nonparametric, semi-parametric, and parametric models, both in abstract terms and in applied contexts such as risk and altruism.

Another important strand of literature concerns the analysis of Random Utility Models (RUMs) in consumer settings. These works aim to simplify the empirical analysis of RUMs.<sup>7</sup> Although there are exceptions, such as Hoderlein and Stoye (2014, 2015), most of the literature focuses on general distributions of types, which are not necessarily ordered. A central message from this literature, pioneered by Kitamura and Stoye (2018), is that determining stochastic rationalizability in a demand setting can be accomplished by analyzing a finite set of points from the cumulative distribution functions that describe the data.<sup>8</sup> The technical details in Theorem 1, along with the applied results in Corollaries 2 and 4, and the political economy example discussed in Section 3.1, shed light on the possibility of extending this principle to other settings. In essence, the finite number of checks arises from the type of rationalizability used in consumer settings (e.g., via monotone and convex preferences) and the specific structure of linear budget sets. This principle does not extend to other menus and utilities. As we discuss in Section 3.1 using the case of political domains with single-peaked preferences, a finite number of checks may be insufficient when menus overlap. Furthermore, even with linear budget sets, if more structured utilities are considered (e.g., expected utilities in risk domains), the linkage of marginal utilities across all possible consumption bundles and budget sets requires a comprehensive analysis of quantiles. However, even in cases where all quantiles must be analyzed, ORUMs retain computational tractability due to the ordered structure of utilities and choices. We contribute to this literature by emphasizing this important aspect of the problem and providing new characterization results.

### 3. NONPARAMETRIC ORUMS

We focus on a setting involving linear, continuous, decision problems for three key reasons. First, while ubiquitous in applications, the theoretical foundations of models adopting this setting remain under-explored in the stochastic choice literature. Second, the continuous structure facilitates theoretical treatment and makes the results easier

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<sup>7</sup>Recent papers bridging the gap between choice-based foundations and the econometric implementation of stochastic models include Dardanoni, Manzini, Mariotti, and Tyson (2020), Aguiar and Kashaev (2021), Apesteguia and Ballester (2021), and Kovach and Tserenjigmid (2022).

<sup>8</sup>See Kashaev, Aguiar, Plávala, and Gauthier (2023) for the application of this technique to the study of dynamic RUMs.

to interpret. Third, as demonstrated in the applications in Section 6, our results apply directly to various classic economic applications involving choices in linear budget sets. Nevertheless, our analysis extends readily to other settings involving non-linear menus, as briefly discussed in Section 7, or discrete menus, as we elaborate in Appendix B.

Let  $X \subseteq \mathbb{R}^K$  be a convex space of alternatives. There is a collection  $\{A_j\}_{j=1}^J$  of decision problems, or menus, that are ordered line segments of  $X$ . That is, each menu  $A_j$  consists of two corner alternatives,  $\underline{x}_j$  and  $\bar{x}_j$ , and their convex combinations, i.e.,

$$A_j = \{(1 - a)\underline{x}_j + a\bar{x}_j : \underline{x}_j, \bar{x}_j \in X \text{ and } a \in [0, 1]\}.$$

Thus, any alternative  $x \in A_j$  is determined by its relative position in the line segment,  $a_j(x) \in [0, 1]$ , which is the unique value such that

$$x = (1 - a_j(x))\underline{x}_j + a_j(x)\bar{x}_j.$$

Using the relative position of alternatives in menus, we assume that choice data corresponds to a collection  $F = \{F_j\}_{j=1}^J$  of cumulative distribution functions (CDFs) over the interval  $[0, 1]$ . That is, given menu  $A_j$  and value  $a \in [0, 1]$ , the value  $F_j(a)$  describes the choice mass of alternatives  $x$  in menu  $A_j$  for which  $a_j(x) \leq a$ . We assume that each CDF is continuous on  $[0, 1)$  and strictly increasing.<sup>9</sup>

To discuss ORUM-rationalizability, let  $\mathcal{U}$  denote the class of relevant utility functions on  $X$ . Assume for now that  $\mathcal{U}$  is given by all utility functions that produce a unique maximizer in each menu. Given a utility function  $U \in \mathcal{U}$ , let  $a_j(U)$  represent the  $a$ -value of the unique alternative in menu  $A_j$  that maximizes utility according to  $U$ .<sup>10</sup> An ORUM has two components:

- (1) Ordered-choice: The model is based on an ordered set of latent types (or simply types), represented by  $\mathbb{R}$ , with each type associated with a utility function such that higher types select higher alternatives. Formally, there is a *type-utility map*  $\gamma : \mathbb{R} \rightarrow \mathcal{U}$  such that, for every menu  $A_j$ , the map of maximizers  $a_j(\gamma(t))$  is continuous and increasing in  $t$ . In alignment with our assumption on data, we work with maps  $\gamma$  that generate maps of maximizers that are strictly increasing over the (menu-dependent) set of interior types, i.e., those leading to  $a \in (0, 1)$ . Consequently, for every menu  $A_j$  and every  $a \in (0, 1)$ , there exists a unique type, denoted by  $t_j^\gamma(a)$ , for which  $a_j(\gamma(t_j^\gamma(a))) = a$ .<sup>11</sup> We denote the class of all such maps as  $\Gamma$ .

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<sup>9</sup>Notice that this allows for mass at any of the corner points of the menu, as commonly observed in many applications. Continuity is a standard technical assumption, and strict monotonicity is analogous to the usual positivity assumption in discrete choice models.

<sup>10</sup>Expressions such as  $a_j(x)$  and  $a_j(U)$  will be used interchangeably, though there is no risk of confusion. The former refers to the position of an alternative  $x$  in menu  $A_j$ , while the latter refers to the position of the alternative that maximizes utility  $U$  in the same menu. As a result,  $a_j(x) = a_j(U)$  conveniently captures the idea that  $x$  is the maximizer of utility  $U$  in menu  $A_j$ .

<sup>11</sup>That is,  $t_j^\gamma$  is just the inverse of the map  $a_j \circ \gamma$ , which is bijective on the set of interior alternatives.

- (2) Stochasticity: The type making choices is subject to randomness. Formally, there is a *type distribution* in the form of a CDF  $G : \mathbb{R} \rightarrow [0, 1]$ . We assume that  $G$  is continuous and strictly increasing, with  $\lim_{t \rightarrow -\infty} G(t) = 0$  and  $\lim_{t \rightarrow \infty} G(t) = 1$ . We denote the class of such CDFs as  $\mathcal{G}$ .

In each menu, the distribution of choices is generated by randomizing over the set of types according to  $G$ , and then maximizing according to the associated utilities given by  $\gamma$ . Formally, we say that data  $F$  is ORUM-rationalizable with type-utility map  $\gamma$  and type distribution  $G$  whenever, for every menu  $A_j$  and every  $a \in (0, 1)$ ,

$$F_j(a) = G(t_j^\gamma(a)).$$

That is, for every menu and interior alternative, the observed cumulative choice mass at that alternative coincides with the mass of types that maximize below it. Given this, the continuity assumptions guarantee that the mass observed at the corners matches the mass of types maximizing at those corners, too. ORUMs clearly represent a restriction of the more general RUMs, with the mass distributed over a unidimensional, ordered collection of utilities.

We now provide a simple characterization of data that is ORUM-rationalizable. This property can be expressed in terms of a deterministic property of quantiles. For every probability value  $p \in (0, 1)$  and every menu  $A_j$ , the basic assumptions on data guarantee that there is a unique alternative  $c_j^p \in A_j$  such that

$$c_j^p = \arg \min_{x: F_j(a_j(x)) \geq p} a_j(x).$$

The choice function  $c^p = \{c_j^p\}_{j=1}^J$  is called the  $p$ -quantile choice function, and, following standard deterministic notions, we say that the  $p$ -quantile choice function  $c^p$  is acyclical if for every collection  $(A_{j_1}, c_{j_1}^p; \dots; A_{j_K}, c_{j_K}^p)$  such that  $c_{j_{k+1}}^p \in A_{j_k}$ ,  $k \in \{1, \dots, K-1\}$ , it holds that  $c_{j_1}^p \notin A_{j_K}$ . In other words, the concatenation of revealed preferences at quantile  $p$  produces no cycle. We say that  $F$  satisfies Quantile Acyclicity whenever every  $c^p$  is acyclical.

**Theorem 1.**  *$F$  is ORUM-rationalizable if and only if  $F$  satisfies Quantile Acyclicity.*

ORUM-rationalizability is thus equivalent to each  $p$ -quantile choice function being acyclical. The intuition for the sufficiency part is as follows: for every  $p \in (0, 1)$ , the acyclicity of the  $p$ -quantile choice function  $c^p$  allows us to identify a utility function  $U^p$  that rationalizes  $c^p$ . Choices must increase across different levels of  $p$  due to the quantile definition of  $c^p$ . Consequently, data can be explained by randomizing uniformly over these quantile utilities. To construct the type-utility map  $\gamma$  and the type distribution  $G$ , we project the interval  $(0, 1)$  onto the real line using a bijection, and consider the corresponding induced distribution over the reals. We do this via the standard logistic transformation. Other transformations could lead to different parameterizations, and we expand on this in Section 4.

**3.1. A political economy example.** We propose a simple political economy example in order to illustrate Theorem 1, the practical testing of the characterizing property, the comparison of ORUMs with the prominent Luce model, and a discussion on the handling of substitution patterns by the model.

Suppose that  $X = \mathbb{R}$ , and let  $\mathcal{U}$  be the class of strictly quasi-concave utility functions on  $X$ . Following the analysis of Moulin (1984), menus will be closed intervals of the real line. ORUM-rationalizability requires the consideration of a type-utility map in which types are assigned strictly quasi-concave utilities, with the family being continuous and strictly increasing in the peaks. When a type distribution is assumed, this induces a distribution of peaks, which is all that matters for choices.

An important aspect of this application, as in many other economic settings, is that the family  $\mathcal{U}$  is assumed to have some minimal structural property. For example, monotonicity in consumption settings, independence in risk settings, or strict quasi-concavity in the current example. Naturally, this requires strengthening the revealed preference notion that defines Quantile Acyclicity. In the present context, strict quasi-concavity implies that, when considering the revealed preference at quantile  $p$ , we learn more than the usual  $c_j^p$  is preferred to every  $x \in A_j$ . Specifically, if  $c_j^p \neq \bar{x}_j$ , we infer the ranking of all alternatives greater than or equal to  $c_j^p$ , with a similar reasoning for alternatives below  $c_j^p$  whenever  $c_j^p \neq \underline{x}_j$ . This revealed information should be incorporated into the acyclicity analysis. With this in mind, the rest follows: the rationalizability of  $c^p$  requires the absence of cycles in the concatenation of these strengthened revealed preferences, and ORUM-rationalizability is equivalent to the corresponding version of Quantile Acyclicity.

Now, suppose that choice data is generated by a uniform distribution of peaks on  $(0, 1)$ , with no mass outside this interval. Consider two menus,  $A_1 = [0, 1]$  and  $A_2 = [0, \frac{1}{2}]$ . Trivially, the data generated by the ORUM is uniform on  $A_1$ , with no mass at the corners. When considering menu  $A_2$ , every type with a peak above  $\frac{1}{2}$  selects the alternative  $\bar{x}_2 = \frac{1}{2}$ , implying that a mass of  $\frac{1}{2}$  is uniformly distributed in the interior, and a mass of  $\frac{1}{2}$  is observed at the upper corner. To see the role of our characterizing property in this setting, note that with two menus related by inclusion, Quantile Acyclicity simply implies asymmetry, i.e., the smaller menu cannot contradict the revelations of the larger one. In particular, when  $p \in (0, \frac{1}{2})$ , since  $c_1^p = p \in A_2 \subseteq A_1$ , we must also have  $c_2^p = p$ . As a result, for menu  $A_2$ , the mass of choices in  $I_1 = [0, p]$  must be  $p$ . By considering  $p$  approaching  $\frac{1}{2}$ , the corner mass described above is derived.

Quantile Acyclicity requires every quantile to be rationalizable. Depending on the application, it may be sufficient to consider a finite number of checks. This is the case, for example, in linear budget sets involving consumption or altruism, as we discuss in Section 6.2. Yet, it can be seen that this feature results from the non-overlapping nature of linear budget sets and the family of utilities at stake. The present example shows that finite checks are not sufficient in some cases. Essentially, ORUM-rationalizability

requires that the revealed distribution of peaks in  $(0, \frac{1}{2})$  is exactly the same for  $A_1$  and  $A_2$ , a feature that cannot be tested with a finite number of checks on  $F_1$  and  $F_2$ . The analysis of expected utility in Section 6 constitutes another example where a finite number of quantiles is insufficient.

We now compare ORUMs to the best-known restriction of RUMs, the Luce model. A continuous version of the Luce model producing uniform choices in  $A_1$  with no mass at the corners requires all alternatives to be indifferent. Hence, contrary to our example, one should also observe uniform choices in  $A_2$  with no mass at the corners. From the point of view of the characterizing property of the Luce model, consider the continuous version of IIA, where the ratio of choice probabilities between two intervals does not depend on the menu to which they belong. IIA then requires that the ratio of choice probabilities for intervals  $I_1 = [0, p]$  and  $I_2 = [p, \frac{1}{2}]$  must be the same in both menus. That is, for menu  $A_2$ , the ratio of mass of choices in  $I_1$  over  $I_2$  should also be  $\frac{p}{\frac{1}{2}-p}$ , and thus the mass of choices belonging to  $I_1$  should be equal to  $\frac{p}{p+\frac{1}{2}-p} = 2p$ , leading to the uniform  $F_2$  with no mass at the corners.

Importantly, note that ORUM is built on the basis of mass over ordered types, which endogenously embeds the ordered substitution patterns contained in the space of alternatives and the family of utilities under consideration. In contrast to the Luce model, in ORUMs, the mass of the removed alternatives  $[\frac{1}{2}, 1]$  is not uniformly distributed among the rest but follows the substitution pattern induced by the ordered structure of the real line and the strictly quasi-concave utilities of types. When these high alternatives are removed, high types concentrate on the highest available alternative in  $A_2$ . By ignoring the ordered structure underlying the political application, the Luce model struggles to accommodate these patterns.

#### 4. SEMI-NONPARAMETRIC ORUMS

Empirical work usually involves a variety of parametric assumptions, either through the adoption of a specific type-utility map or by working with a particular class of type distributions. These assumptions may, for example, facilitate computational estimation or aid in interpreting results. In this section, we briefly discuss the potential implications of restricting each of these channels separately. We first consider the case in which the dimensionality of the type distribution is reduced.<sup>12</sup> The next result follows directly from Theorem 1 and shows that such restrictions alone have no further empirical implications.

**Corollary 1.** *Let  $G \in \mathcal{G}$ .  $F$  is ORUM-rationalizable with type distribution  $G$  if and only if  $F$  satisfies Quantile Acyclicity.*

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<sup>12</sup>Notice that Corollary 1 below fixes a given type distribution. Since this is inconsequential, the theorem extends to cases where the type distribution is assumed to belong to a particular family of distributions, such as in ordered-logit or ordered-probit models.

In other words, parametric restrictions on the type distribution alone have no empirical content per se. The reason for this is that an appropriate relabelling of utilities allows for free modification of the type distribution structure. Thus, Corollary 1 can be viewed as a basic complementary result to Theorem 1, akin to the standard practice of normalization. Let us illustrate this technique using Matzkin (2003), where the class of econometric models of the form  $Y = m(X, \alpha)$ , where  $X$  and  $Y$  are observables,  $\alpha$  is a non-additive unobserved error, and  $m$  is monotone in  $\alpha$ , is studied. Note how, by considering how the observable choice  $Y$  depends on the observable menu of alternatives  $X$  through the unobserved latent variable  $\alpha$  and the optimization process  $m$ , it becomes clear that ORUMs belong to this class of econometric models. For these models, Lemma 1 in Matzkin (2003) demonstrates that the distribution of errors can be fixed without loss of generality, allowing for a number of possible normalizations before further analysis; the same type of logic is precisely the reasoning behind why Quantile Acyclicity is also sufficient for ORUM-rationalizability, even after fixing the type distribution. Thus, Corollary 1 can be viewed as a direct combination of our Theorem 1 and the standard logic of normalization.<sup>13</sup>

Next, we consider the case in which the type-utility map is fixed. For example, in the study of decisions under risk, let  $\mathcal{U}$  represent the general class of expected utilities. An analyst interested in describing risk aversion levels in a population might fix the type-utility map such that  $\gamma(t)$  corresponds to expected utility with a constant relative risk aversion coefficient equal to  $t$ .<sup>14</sup> This modelling assumption is likely to restrict the analyst's flexibility, since the sub-class  $\gamma(\mathbb{R})$  of utilities within the image of  $\gamma$  has a lower dimension than  $\mathcal{U}$ . For instance, CRRA utilities may fail to capture across-menu choice patterns that can only be explained by other expected utility functions. However, the logic presented in Theorem 1 can be directly applied to this semi-nonparametric case by appropriately adapting the required revealed preference property on each quantile of choices to capture the rationalizability by utilities in  $\gamma(\mathbb{R})$ . That is, instead of applying the acyclicity condition, that is relevant when all utilities are considered, we must strengthen the deterministic concept to apply precisely to the sub-class of utilities  $\gamma(\mathbb{R})$ . In our abstract setting, we refer to this as Quantile  $\gamma(\mathbb{R})$ -Rationalizability. In Section 6 we examine the CRRA expected utility case in detail, and in Section 7 we provide another example in the context of altruism with CES utilities.

**Theorem 2.** *Let  $\gamma \in \Gamma$ , with image  $\gamma(\mathbb{R})$ .  $F$  is ORUM-rationalizable with type-utility map  $\gamma$  if and only if  $F$  satisfies Quantile  $\gamma(\mathbb{R})$ -Rationalizability.*

Theorem 2 shows that, modulo the restriction imposed by the assumed sub-family of utilities, the specific parametrization chosen does not result in any loss of generality.

<sup>13</sup>Note that the results in our paper are not primarily concerned with identification. Instead, our goal is to provide necessary and sufficient conditions for all cases, thereby demonstrating the exact empirical content of these models.

<sup>14</sup>The CRRA monetary utility function with coefficient  $t$  is of the form  $\frac{z^{1-t}}{1-t}$  for  $t \neq 1$  and  $\log z$  for  $t = 1$ , with  $z > 0$ .

Once again, we can adapt to any specific labelling of these utilities by finding the exact type distribution that fits this labelling.

Before concluding, we note that empirical work often involves restrictions on both components of the model; that is, the analyst typically fixes a specific type-utility map and imposes some structural property on the type distribution. For example, an analyst studying risk aversion might fix the CRRA expected utility map and assume logistic variation. The combination of these two restrictions generates stronger implications, which we illustrate using the political economy example discussed earlier. This issue is discussed in greater detail in Section 5.

**4.1. A political economy example.** We illustrate the semi-nonparametric results using, again, our political economy example involving the real line and the class of strictly quasi-concave utilities. For the purposes of this section, we simplify the rationalizability question by assuming that data is observed over a single interval  $A_1 = [-k, k]$ . Notice that, since there is only a single menu, Quantile Acyclicity for single-peaked preferences trivially holds, and thus any data is ORUM-rationalizable. In particular, suppose that  $F_1(0) = \frac{1}{3}$ ,  $F_1(\frac{1}{2}) = \frac{1}{2}$ , and  $\lim_{a \rightarrow 1} F_1(a) = \frac{4}{5}$ .<sup>15</sup> ORUM-rationalizability can be achieved with any continuous and strictly increasing type-utility map and type distribution for which the induced CDF of peaks has value  $\frac{1}{3}$  at  $-k$ ,  $\frac{1}{2}$  at 0, and  $\frac{4}{5}$  at  $k$ .

Corollary 1 informs us that we can account for logistic variation of the latent variable by selecting an appropriate type-utility map. For example, define  $\gamma(t)$  as the utility function  $-(x - f(t))^2$ , where  $f(t) = \frac{t}{2}$  whenever  $t \geq 0$ , and  $f(t) = t$  whenever  $t < 0$ . If we then select a logistic distribution with location and scale parameters set to 0 and 1, respectively, this model rationalizes  $F_1$ .

Theorem 2 shows that fixing the type-utility map  $\gamma$  may require an adaptation of the quantile rationalizability concept. Suppose, for example, that we fix the type-utility map where type  $t$  is assigned the Euclidean utility  $E(t)$ , defined by  $-(x - t)^2$ . Euclidean utilities form a sub-family of strictly quasi-concave utilities, and additional revelations are possible because, for example, observing an interior choice reveals the ranking of all alternatives in  $\mathbb{R}$ . In such cases, we may need to define a deterministic notion of Euclidean rationalizability. However, with choices over intervals, all that matters is the distribution of peaks, and thus the Euclidean restriction becomes inconsequential; the same deterministic choice behaviors can still be rationalized. In particular, our example with a single menu remains rationalizable, and to explain the data using  $E$ , we simply need to select a type distribution with CDF values at  $-k$ , 0, and  $k$ , as discussed above. The combination of  $E$  and the constructed type distribution rationalizes the data.

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<sup>15</sup>For the present discussion, it suffices to consider the values of  $F_1$  at a few points. Therefore, we will show the rationalization only at these specific points.

Now, consider a scenario where both the specific type-utility map  $E$  and a restriction to logistic type distributions are imposed. Under these assumptions, the data  $F_1$  is no longer ORUM-rationalizable. Notice that, given  $F_1(\frac{1}{2}) = \frac{1}{2}$ , the median value of the latent variable must be 0. Since the peaks of types  $-k$  and  $k$  are symmetric with respect to 0, the logistic assumption would imply that  $F_1(0) = 1 - \lim_{a \rightarrow 1} F_1(a)$ , which results in a contradiction. Thus, parametric assumptions on both components of the model lead to consequences beyond those described in Theorem 2, which will be further analyzed in the next section.

## 5. PARAMETRIC ORDERED-LOGIT MODEL

As illustrated by our previous example, when parametric restrictions are imposed on both the type-utility map and the type distribution, the model becomes restricted in ways that cannot be easily captured by Theorem 1 or the adjustment proposed in Theorem 2. It is therefore evident that the foundations of parametric models often used in empirical work, such as ordered-logit models, require more sophisticated properties that restrict data across quantiles. In this section, we take an initial step toward describing the exact empirical content of such models. Specifically, we study parametric ordered-logit models, a popular tool in economics as well as other disciplines such as political science, sociology, and biology.<sup>16</sup>

To simplify the analysis, we assume the following richness condition: for any two menus  $A_j$  and  $A_{j'}$ , there exists a sequence of menus  $A_j = A_{j_0}, \dots, A_{j_k}, \dots, A_{j_K} = A_{j'}$  such that, for each  $k \in \{0, \dots, K-1\}$ , there exists an interval of types that produce interior maximizers in both  $A_{j_k}$  and  $A_{j_{k+1}}$ . To motivate this assumption, note that if at least one of the corner alternatives has no mass, the richness assumption is trivially satisfied.<sup>17</sup> However, if choices at both corners are relevant, the richness condition only requires that the intervals of utilities generating interior maximizers overlap weakly, after connecting a chain of menus. For example, in a consumption setting, any pair of menus should be connected by other menus (likely involving intermediate prices) to produce the desired overlap.

We now study the parametric ordered-logit model, in which a generic type-utility map  $\gamma : \mathbb{R} \rightarrow \mathcal{U}$  is fixed, and the type distribution is restricted to the logistic family, denoted  $\mathcal{G}^L$ . Recall that a logistic type distribution takes the form

$$G^{(\tau, \sigma)}(t) = \frac{1}{1 + e^{-(t-\tau)/\sigma}},$$

where  $\tau \in \mathbb{R}$  and  $\sigma > 0$  are the location and scale parameters.

We now present the key properties of this ordered-logit model, which capture the requirements on corner and interior alternatives that arise when a logistic distribution

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<sup>16</sup>For references to the diversity of economic applications of the ordered-logit model, see the discussion in footnote 1 of the Introduction.

<sup>17</sup>This feature is present in the two applications discussed in Section 6.

is applied to the given type-utility map  $\gamma$ . The requirement on the corner alternatives follows from the fundamental properties of Random Utility Models (RUMs). In this continuous ordered setting, the existence of a single type leading to a corner implies the existence of an unbounded interval of types with the same property, resulting in a strictly positive mass for this corner.<sup>18</sup>

**Corner Extremeness (CE).**  $F_j(0) > 0$  (respectively,  $\lim_{a \rightarrow 1} F_j(a) < 1$ ) if and only if there exists  $t \in \mathbb{R}$  such that  $a_j(\gamma(t)) = 0$  (respectively,  $a_j(\gamma(t)) = 1$ ).

To formulate our second property, consider the cumulative log-odds of any interior value  $a \in (0, 1)$ , defined as

$$\ell_j(a) = \log \left( \frac{F_j(a)}{1 - F_j(a)} \right).^{19}$$

It is well known that when data is generated by a logistic distribution with parameters  $(\tau, \sigma)$ , the cumulative log-odds  $\ell_j(a)$  correspond to the standardized type with  $a$  as the maximizer, i.e.,  $\ell_j(a) = \frac{t_j^\gamma(a) - \tau}{\sigma}$ . Now, consider two menus and a pair of non-corner alternatives from each menu. If the sum of the types associated with the first pair of alternatives equals the sum of the types associated with the second pair, then the sum of the cumulative log-odds of both pairs should also coincide.<sup>20</sup>

**Cumulative Log-odds Additivity (CLA).** Let  $a, b, a', b' \in (0, 1)$  and  $A_j, A_{j'}$  be such that  $t_j^\gamma(a) + t_j^\gamma(b) = t_{j'}^\gamma(a') + t_{j'}^\gamma(b')$ . Then,  $\ell_j(a) + \ell_j(b) = \ell_{j'}(a') + \ell_{j'}(b')$ .

Note that when  $a = b$  and  $a' = b'$ , CLA implies that the same type must have the same associated log-odds, and hence the same quantile, in every menu. This implies that quantiles are rationalized by utilities in  $\gamma(\mathbb{R})$ . Importantly, CLA not only imposes this requirement but also constrains the sum of log-odds for pairs of alternatives.

Theorem 3 shows that these two basic properties, CE and CLA, are not only necessary but also sufficient for data to be ORUM-rationalizable with type-utility map  $\gamma$  and a type distribution in  $\mathcal{G}^L$ .

**Theorem 3.** *Let  $\gamma$  be any type-utility map. The data  $F$  is ORUM-rationalizable with type-utility map  $\gamma$  and a type distribution in  $\mathcal{G}^L$  if and only if  $F$  satisfies CE and CLA.*

The proof of the sufficiency part of Theorem 3 involves several steps. First, when considering the interior alternatives in a given menu, the data immediately induces a specific CDF over the corresponding interval of types. When relevant, we account for the censoring produced by corner choices, which are optimal for an unbounded interval of types. The masses observed at the corners must be appropriately distributed among all rationalizing types, guaranteed to exist by CE, in such a way that the constructed

<sup>18</sup>This is akin to the extremeness property described by Gul and Pesendorfer (2006).

<sup>19</sup>This is also called the logit function, which is the inverse of the logistic distribution.

<sup>20</sup>Since  $\gamma$  is fixed, the mapping from types to alternatives is known, and CLA can thus be verified.

CDF over all types satisfies the additivity requirement imposed by CLA for interior alternatives. We address this using a recursive construction. Second, the ordered-logit functional form requires us to build upon Theorem 2.1.5 of Galambos and Kotz (1978). This classical statistical result provides a necessary and sufficient condition for a single CDF over the real line, that has been assumed to be symmetric about the origin, to be logistic. We extend this result to our revealed preference setting, where (i) distributions may have arbitrary means and are not proven to be symmetric, and (ii) we have a collection of menu-dependent distributions, not yet shown to coincide. Our CLA property proves sufficient to demonstrate that these menu-dependent distributions are all logistic and share the same location and scale parameters. Finally, it is important to note that the parameters  $(\tau, \sigma)$  of the logistic type distribution that rationalize the data must be unique.

**5.1. A political economy example.** The properties of CE and CLA may seem abstract, but this is primarily because the type-utility map  $\gamma$  has been presented abstractly. Let us return to the parameterized ordered-logit model using the Euclidean map  $E$ . In Section 4.1, we learned that this model restricts data, even with a single menu.<sup>21</sup> With respect to CE, notice that for any menu  $A_j = [\underline{x}_j, \bar{x}_j]$ , the corner alternatives are always the maximizers for some type (specifically, for types with peaks below  $\underline{x}_j$  and above  $\bar{x}_j$ , respectively). Consequently, CE reads as (for any menu):

$$[\text{CE}_{PE}] \quad F_j(0) > 0 \quad \text{and} \quad \lim_{a \rightarrow 1} F_j(a) < 1.$$

Regarding CLA, notice that with Euclidean preferences, the utility maximized at  $x$  is the one with a peak at  $x$ , and hence  $t_j^E(a_j(x)) = x$ . Therefore, CLA reads as follows (for any menus and interior alternatives):

$$[\text{CLA}_{PE}] \quad x + y = x' + y' \Rightarrow \ell_j(a_j(x)) + \ell_j(a_j(y)) = \ell_{j'}(a_{j'}(x')) + \ell_{j'}(a_{j'}(y')).$$

Note that our analysis can be applied to the study of other type distributions.<sup>22</sup> When considering any other family of continuous and strictly increasing distributions, CE would still apply, while the CLA requirement should be modified to capture the structural implications of the corresponding statistical distribution. For instance, if we consider a Gaussian latent variable, the standardized type  $\frac{t_j^G(a) - \tau}{\sigma}$  would now take the form  $\Phi_j^{-1}(a)$ , where  $\Phi_j^{-1}$  denotes the inverse of the CDF of the standard normal. Setting aside the lack of a closed-form solution for this expression, one could operationalize a version of the CLA property by simply replacing  $\ell_j(a)$  with  $\Phi_j^{-1}(a)$ .

Moreover, the structure of the CLA property also helps in deciding which family of type distributions to work with. An analyst can start by fixing the type-utility map  $\gamma$ ,

<sup>21</sup>Indeed, we showed how data should be symmetric around the alternative occupying the median quantile, a property implied by the analysis here.

<sup>22</sup>An alternative approach is to consider mixtures of the logistic distribution. This can be seen as a parallel exercise to the mixed-logit analysis. In Appendix C, we show that any ORUM with a given type-utility map  $\gamma$  can be approximated by a sequence of mixed ordered logits using the same  $\gamma$ .

understanding the implications of this assumption as outlined in Theorem 2. Then, if a parametric exercise is performed, an analysis of the changes in the sums of log-odds,  $\Phi^{-1}$ , or the corresponding expression given by the distribution at stake, may guide the selection of the most appropriate type distribution.

## 6. APPLICATIONS

An advantage of our results is that they are portable to the analysis of specific economic settings of interest. In this section, we stress the applicability of our framework by analyzing in detail two influential settings, one involving risk and the other altruism, that have recently attracted a great deal of attention, both theoretically and experimentally. In both cases, the analyst can observe a number of choices using the, by now standard, methodology of linear budget sets.

As illustrated in Section 5.1 with the political economy example, when a specific setting is adopted, all properties can be expressed in terms of the appropriate fundamentals and take on a form that is familiar to the expert, allowing for direct empirical examination. In this section, we particularize Theorems 1 to 3, providing nonparametric, semi-nonparametric, and parametric characterizations for the settings of risk and altruism. In Appendix D, we connect our theoretical results further to applied work by discussing their application to standard datasets and some initial ideas on testing.

**6.1. Application: Risk.** Let  $X = [0, 1] \times \mathbb{R}_+^2$  represent the set of all possible (two) state-contingent lotteries.<sup>23</sup> A menu is a collection of lotteries in which all have the same state probabilities, and their payouts belong to a linear budget set. Formally, wealth is equal to 1, and let  $\pi_j^i > 0$  denote the price of allocating money to state  $i$  in menu  $A_j$ . Then, a menu is the collection of all affordable lotteries of the form  $(q_j^1; x^1, x^2)$ , where  $q_j^1 \in [0, 1]$  describes the menu-dependent probability of state 1 (hence,  $q_j^2 = 1 - q_j^1$  describes the probability of state 2), and  $(x^1, x^2) \in \mathbb{R}_+^2$  are the possible payout combinations. With the usual budget set notation, the affordable lotteries are  $B_j = \{(q_j^1; x^1, x^2) \in X : \pi_j^1 x^1 + \pi_j^2 x^2 \leq 1\}$ .<sup>24</sup> We assume, without loss of generality, that state 1 pays more in expectation, i.e.,  $\phi_j^1 \equiv \frac{\pi_j^1}{q_j^1} < \frac{\pi_j^2}{q_j^2} \equiv \phi_j^2$ , and denote  $\phi_j = \frac{\phi_j^2}{\phi_j^1}$ . Given the assumptions on utility laid down below, potential maximizers belong to the line segment  $A_j$  between the corner allocation  $\underline{x}_j = (q_j^1; \frac{1}{\pi_j^1}, 0)$  and the equal-payout allocation  $\bar{x}_j = (q_j^1; \frac{1}{\pi_j^1 + \pi_j^2}, \frac{1}{\pi_j^1 + \pi_j^2})$ . Note that larger levels of risk aversion are intuitively

<sup>23</sup>We discuss the case of two states because ORUMs in general, and ordered-logit models in particular, are uni-dimensional in nature; they create one-to-one maps from a latent variable to the space of choices. We briefly address this point in Section 7. Moreover, as shown in Apesteguia, Ballester, and Gutierrez-Daza (2025), the lessons learned in the uni-dimensional analysis can be transferred via conditional analysis to the study of multi-dimensional settings.

<sup>24</sup>Given a menu, the proper decision variables involve  $x^1$  and  $x^2$ , but we need to keep in the notation the menu constant  $q_j^1$  to fully describe the lottery.

reflected in larger investments in state 2. For this application, it is more convenient to represent each alternative  $x \in A_j$  by its unique associated ratio  $r_j(x) \in [0, 1]$ , given by  $r_j(x) = \frac{x^2}{x^1}$ , rather than by the value  $a_j(x)$  used above.<sup>25</sup>

We begin with the nonparametric and semi-nonparametric analysis. Given the risk setting, we first focus on the class  $\mathcal{U}$  of expected utilities associated with continuous, strictly increasing, and concave monetary utility functions. To illustrate the semi-nonparametric case, we fix the type-utility map *CRRA*, where type  $t$  corresponds to the expected utility with a relative risk aversion coefficient equal to  $t$ .<sup>26</sup>

The nonparametric study of ORUM-rationalizability requires expected utility rationalizability at the quantile level. We can borrow from the deterministic analysis of expected utility rationalizability by Kubler, Selden, and Wei (2014). Following their paper, we say that the quantile choice function  $c^p$  formed by the collection  $\{c_j^p = (q_j^1; c_j^{p,1}, c_j^{p,2})\}_{j=1}^J$  satisfies the Strong Axiom of Revealed Expected Utility (SAREU) whenever, for every sequence of menus  $A_{j_1}, \dots, A_{j_K}$ , it holds that  $L(j_1, j_2) \cdot L(j_2, j_3) \cdot \dots \cdot L(j_{K-1}, j_K) \cdot L(j_K, j_1) < 1$ , where<sup>27</sup>

$$L(j, j') = \begin{cases} 0 & \text{whenever } c_{j'}^{p,s'} > c_j^{p,s} \text{ for all } s, s' \in \{1, 2\}, \\ \max_{s, s': c_j^{p,s} > c_{j'}^{p,s'}} \frac{\phi_j^s}{\phi_{j'}^{s'}} & \text{otherwise.} \end{cases}$$

The simplest violation of SAREU involves a single menu  $A_j$  such that  $L(j, j) > 1$ . This corresponds to  $c_j^{p,2} > c_j^{p,1}$ , i.e., a violation of first-order stochastic dominance (since, as discussed above, larger investments in the state paying more in expectation should be unequivocally observed). SAREU eliminates these and more complex weighted cycles, an issue that will be discussed in detail in the empirical part below. We say that  $F$  satisfies Quantile SAREU whenever every  $c^p$  satisfies SAREU.

Next, consider the semi-nonparametric model. From the standard first-order condition analysis, it is well-known that the relative risk aversion coefficient of the type maximizing at  $x$  in menu  $A_j$  is  $-\frac{\log \phi_j}{\log r_j(x)}$ . Thus, for a choice function to be rationalizable by a CRRA expected utility, this expression should be, for any given quantile, constant across menus. Formally, we say that the quantile choice function  $c^p$  satisfies the Strong Axiom of Revealed CRRA Expected Utility (SARCEU) whenever, for every

<sup>25</sup>Notice that the definition of quantiles is not affected by this ordinal transformation. This is merely more convenient for the exposition of the parametric case.

<sup>26</sup>Recall that, as shown in Corollary 1, the semi-nonparametric model imposing assumptions on the type distribution is equivalent to the nonparametric model, hence we omit its discussion here and in Section 6.2.

<sup>27</sup>As in Kubler, Selden, and Wei (2014), we assume that  $c_j^{p,s} \neq c_{j'}^{p,s'}$  for all  $j, j'$  and  $s, s'$ .

pair of menus  $A_j$  and  $A_{j'}$ , it holds that

$$\frac{\log \phi_j}{\log r_j(c_j^p)} = \frac{\log \phi_{j'}}{\log r_{j'}(c_{j'}^p)}.^{28}$$

We say that  $F$  satisfies Quantile SARCEU whenever every  $c^p$  satisfies SARCEU.

**Corollary 2.** *In the risk domain:*

- (1)  $F$  is ORUM-rationalizable if and only if  $F$  satisfies Quantile SAREU.
- (2)  $F$  is ORUM-rationalizable with type-utility map CRRA if and only if  $F$  satisfies Quantile SARCEU.

We now proceed to study the parametric case. We consider again the parametrization CRRA, and require the type distribution to belong to the class  $\mathcal{G}^L$  of logistic distributions. We apply Theorem 3 to analyze the properties in this setting.

First, notice that the corner allocation  $\underline{x}_j = (q_j^1; \frac{1}{\pi_j^1}, 0)$  corresponds to the menu-independent interval of types  $(-\infty, 0]$ , and must have strictly positive mass. Meanwhile, the corner allocation  $\bar{x}_j = (q_j^1; \frac{1}{\pi_j^1 + \pi_j^2}, \frac{1}{\pi_j^1 + \pi_j^2})$  is sub-optimal for all utilities in the CRRA class and must have zero mass. Therefore, CE reads as follows, for every menu:

$$[\text{CE}_R] \quad F_j(0) > 0 \quad \text{and} \quad \lim_{r \rightarrow 1} F_j(r) = 1.$$

Since the type maximizing at  $r \in (0, 1)$  is given by  $-\frac{\log \phi}{\log r}$ , the CLA condition for any pair of menus and interior ratios is:

$$[\text{CLA}_R] \quad \log \phi_j \left( \frac{1}{\log r} + \frac{1}{\log s} \right) = \log \phi_{j'} \left( \frac{1}{\log r'} + \frac{1}{\log s'} \right) \Rightarrow \ell_j(r) + \ell_j(s) = \ell_{j'}(r') + \ell_{j'}(s').$$

Corollary 3 follows directly from these conditions:

**Corollary 3.** *In the risk domain,  $F$  is ORUM-rationalizable with type-utility map CRRA and a type distribution in  $\mathcal{G}^L$  if and only if  $F$  satisfies  $\text{CE}_R$  and  $\text{CLA}_R$ .*

In relation to the comparison between the parametric and semi-nonparametric properties, note that the quantile version of SARCEU requires  $\frac{\log \phi_j}{\log r} = \frac{\log \phi_{j'}}{\log r'} \Leftrightarrow \ell_j(r) = \ell_{j'}(r')$ . This follows readily from  $\text{CLA}_R$  by considering the case in which  $r = s$  and  $r' = s'$ .  $\text{CLA}_R$  strengthens SARCEU by requiring this principle to apply across pairs of quantiles.

**6.2. Application: Altruism.** Let  $X = \mathbb{R}_+^2$  represent the set of all possible monetary allocations in which the first component refers to the payment to oneself and the second refers to the payment to another person. Individuals confront linear budget sets of the form  $B_j = \{x = (x^1, x^2) \in X \mid \pi_j^1 x^1 + \pi_j^2 x^2 \leq 1\}$ , where  $\pi_j^i > 0$  denotes the cost of allocating money to person  $i$  in budget set  $B_j$ . Below we assume monotonicity in preferences, leading choices to belong to the line segment, denoted  $A_j$ , between the

<sup>28</sup>When the choice is such that  $c_j^{p,2} = 0$ , the expression should be read as 0.

corner allocations  $\underline{x}_j = (\frac{1}{\pi_j^1}, 0)$  and  $\bar{x}_j = (0, \frac{1}{\pi_j^2})$ . Note that larger levels of altruism are reflected in larger transfers to individual 2. In this application, it is convenient to use the following parametrization of the allocations in a menu  $A_j$ :  $v_j(x) \in [0, 1]$  given by  $v_j(x) = 1 - e^{-r_j(x)}$ .<sup>29</sup> Also, denote  $\pi_j = \frac{\pi_j^2}{\pi_j^1}$ .

We consider first the nonparametric analysis. In line with standard models of social preferences, we start by considering the class  $\mathcal{U}$  of utilities that are continuous and either: (i) strictly monotone in the first component (and either constant or decreasing in the second component), or (ii) strictly increasing in both components and strictly convex. The characterization of this case will be based on the classical Weak Axiom of Revealed Preference (WARP). In our setting, the quantile choice function  $c^p$  satisfies WARP whenever for every pair of menus  $A_j, A_{j'}$  such that  $c_j^p \in B_{j'}$  and  $c_{j'}^p \in B_j$ , it is  $c_j^p = c_{j'}^p$ . As it is well-known, this is basically the asymmetry part of the Acyclicity property of Section 3, together with monotonicity. We say that  $F$  satisfies Quantile WARP whenever every  $c^p$  satisfies WARP.

In the semi-nonparametric case, we consider CES-type utility functions  $(x^1)^\alpha + t(x^2)^\alpha$ , where  $\alpha \in (0, 1)$  is a constant, and allow for variation in the altruism coefficient  $t \in \mathbb{R}$ . The study of this case follows by applying Theorem 2 in combination with the following deterministic analysis. From the standard first-order condition, the altruism coefficient is equal to  $\pi_j r_j(x)^{1-\alpha}$ , where recall that  $r_j(x) = \frac{x^2}{x^1}$ . For a given quantile, the altruism coefficient should be equal across pairs of menus. Hence, for any given quantile  $p$ , this allows to obtain  $\alpha$  from two menus as  $\frac{\log(\pi_j/\pi_{j'})}{\log(r_j(c_j^p)/r_{j'}(c_{j'}^p))} + 1$ . We then say that the collection of choices  $c^p$  satisfies the Weak Axiom of Revealed CES Preference (WARCESP) whenever for every three menus  $A_j, A_{j'}, A_{j''}$ , either  $c_j^{p,2} = 0$  always holds or

$$0 < \frac{\log(\pi_j/\pi_{j'})}{\log(r_j(c_j^p)/r_{j'}(c_{j'}^p))} + 1 = \frac{\log(\pi_j/\pi_{j''})}{\log(r_j(c_j^p)/r_{j''}(c_{j''}^p))} + 1 < 1.$$

We say that  $F$  satisfies Quantile WARCESP whenever every  $c^p$  satisfies WARCESP for the same constant.

**Corollary 4.** *In the altruism domain,*

- (1)  *$F$  is ORUM-rationalizable if and only if  $F$  satisfies Quantile WARP.*
- (2)  *$F$  is ORUM-rationalizable with type-utility map CES if and only if  $F$  satisfies Quantile WARCESP.*

Corollary 4 is similar in spirit to Corollary 2. The nonparametric case is obtained with a relatively classical and broad set of preferences, while the semi-nonparametric case is derived by using the CES family of utility functions. Importantly, we want to focus our attention on the altruism parameter  $t$ , and not on the curvature  $\alpha$ , so the second part of Corollary 4 requires not only the assumption that every quantile

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<sup>29</sup>Where we set  $v_j(x) = 1$  when  $x^1 = 0$ .

is CES-rationalizable but also that all quantiles are CES-rationalized with the same curvature.

We now proceed to study the parametric case by using the CES map for a given value of  $\alpha$ , and imposing the logistic distribution over the type space  $t$  (or equivalently, over the altruism parameter). The study of rationalizability is again a direct consequence of the application of Theorem 3. To understand how property CE reads in this setting, notice again that for any line segment  $A_j$ , the corner allocation  $\underline{x}_j = (\frac{1}{\pi_j}, 0)$  is associated with a constant interval of CES utilities ( $t \leq 0$ ) while the corner allocation  $\bar{x}_j = (0, \frac{1}{\pi_j^2})$  is never selected. Then, for any menu:

$$[\text{CE}_A] \quad F_j(0) > 0 \text{ and } \lim_{v \rightarrow 1} F_j(v) = 1.$$

To see how property CLA reads, notice that the type that maximizes at the interior alternative  $x$  is the one with altruism coefficient equal to  $\pi_j r_j(x)^{1-\alpha}$ . Then, for any pair of menus and interior alternatives:

$$[\text{CLA}_A] \quad \pi_j \left( \left[ \log \frac{1}{1-v} \right]^{1-\alpha} + \left[ \log \frac{1}{1-w} \right]^{1-\alpha} \right) = \pi_{j'} \left( \left[ \log \frac{1}{1-v'} \right]^{1-\alpha} + \left[ \log \frac{1}{1-w'} \right]^{1-\alpha} \right) \Rightarrow$$

$$\ell_j(v) + \ell_j(w) = \ell_{j'}(v') + \ell_{j'}(w').$$

We omit the proof of the following immediate result.

**Corollary 5.** *In the altruism domain,  $F$  is ORUM-rationalizable with type-utility map CES and a type distribution in  $\mathcal{G}^L$  if and only if  $F$  satisfies  $\text{CE}_A$  and  $\text{CLA}_A$ .*

## 7. FINAL REMARKS

This paper investigates the testable implications of commonly imposed parametric assumptions in ORUMs, specifically focusing on the case where menus are represented as linear segments, a structure that aligns well with several important economic applications. Importantly, notice that when a type-utility map is fixed, as in the semi-nonparametric and parametric cases, maximizers form a one-dimensional curve ordered by the type-utility map. As a result, the analysis can be based on these induced ordered curves, and hence, our results apply far beyond the setting and applications considered.<sup>30</sup> In Appendix B, we also extend our results to the case of finite problems, demonstrating the flexibility of the setting.

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<sup>30</sup>In particular, no restriction on the linearity of menus or on the number of goods is of relevance in the semi-nonparametric and parametric cases. For the nonparametric results, it would be critical to examine the conditions under which such one-dimensional curves can be derived.

## APPENDIX A. PROOFS

**Proof of Theorem 1:** We start by proving the necessity part. Suppose that data  $F$  is ORUM-rationalizable with type-utility map  $\gamma$  and type distribution  $G$ , and consider any probability value  $p \in (0, 1)$ . Given that  $G$  is continuous and strictly increasing, there is a unique type  $G^{-1}(p)$  for which the cumulative mass is equal to  $p$ , i.e.,

$$G^{-1}(p) = \{t : G(t) = p\} = \min_{t: G(t) \geq p} t.$$

We claim that, for every menu  $A_j$ , the utility associated to type  $G^{-1}(p)$ ,  $\gamma(G^{-1}(p))$ , has maximizer  $c_j^p$ . Suppose, by way of contradiction that this is not the case, i.e., suppose that there exists a menu  $A_j$  for which the maximizer of  $\gamma(G^{-1}(p))$  is an alternative  $x$  such that  $a_j(x) \neq a_j(c_j^p)$ . If  $a_j(x) < a_j(c_j^p)$ , the ordered structure of choices generated by utilities guarantees that for every type  $t \leq G^{-1}(p)$ ,  $a_j(\gamma(t)) \leq a_j(x)$ . Since  $(\gamma, G)$  rationalizes data, it must be  $F_j(a_j(x)) \geq G(G^{-1}(p)) = p$ , and  $x$  contradicts the definition of  $c_j^p$ . If  $a_j(x) > a_j(c_j^p)$ , it must be  $c_j^p \neq \bar{x}_j$  and given the assumptions, we know that there is a type  $t^*$  such that  $a_j(\gamma(t)) > a(c_j^p)$  if and only if  $t > t^*$ . It is  $t^* < G^{-1}(p)$  and, given the assumptions on  $G$  and the rationalization of data, the mass below  $c_j^p$  would be strictly lower than  $p$ , a contradiction. We have then proved that the maximizers of type  $G^{-1}(p)$  are described by the choice function  $c^p$ . Given classical results, this choice function must be acyclical, and necessity follows.

We now prove the sufficiency part. Suppose that for a given probability value  $p \in (0, 1)$  the choice function  $c^p$  is acyclical. Given classical results, we can construct a utility function  $U^p \in \mathcal{U}$  that is strictly positive over the finite set of alternatives  $\{c_j^p\}_{j=1}^J$  and such that  $U^p(x) = 0$  for the rest of alternatives in  $X$ , that rationalizes these  $p$ -quantile choices. That is, for every menu  $A_j$ ,  $U^p$  gives  $c_j^p$  as a unique maximizer. Assign, to every type  $t \in \mathbb{R}$ , the utility function  $\gamma(t)$  given by  $U^{\frac{1}{1+e^{-t}}}$ . We first claim that, for every menu, the maximizers induced by  $\gamma$  are increasing in  $t$ . To see this, consider any menu  $A_j$  and, given the construction, we merely need to prove that choices  $c_j^{\frac{1}{1+e^{-t}}}$  are increasing in  $t$ , which follows immediately from the quantile definition of such alternatives and the obvious fact that  $F_j$  is non-decreasing. The strictly increasing nature in the interior and the continuity of choices follow similarly. Now, consider  $G$  to be the logistic distribution with location 0 and variance 1. We claim that, for every menu  $A_j$  and every  $a \in (0, 1)$ ,  $F_j(a)$  coincides with the mass of types maximizing below  $a$ . Given the non-decreasing nature of the maximizing alternatives, we need to prove that the utility function  $U^{F_j(a)}$  is the last utility with maximizer below  $a$ . First, consider  $p > F_j(a)$ . Since  $a$  has not reached cumulative probability  $p$ , it must be that  $a(c_j^p) > a$ , and since utility  $U^p$  rationalizes  $c_j^p$ , the maximizer of  $U^p$  lies strictly above  $a$ . Second, consider the utility function  $U^{F_j(a)}$ . By construction,  $c_j^{F_j(a)}$  lies below  $a$  and since  $U^{F_j(a)}$  rationalizes  $c_j^{F_j(a)}$ , the maximizer of  $U^{F_j(a)}$  lies below  $a$ . This concludes the sufficiency part and the proof.  $\blacksquare$

**Proof of Corollary 1:** Necessity follows directly from Theorem 1. For sufficiency, let  $G \in \mathcal{G}$  and define  $U^p$  as in the proof of Theorem 1. Then, for every type  $t \in \mathbb{R}$ , define  $\gamma(t)$  as the utility  $U^{G(t)}$ . That is, type  $t$  is assigned the utility that corresponds exactly to the quantile, according to  $G$ , of this type. The ordered-choice structure of type-utility map  $\gamma$  is immediate and ORUM-rationalizability follows from reproducing the proof of Theorem 1 with the pair  $(\gamma, G)$ . ■

**Proof of Theorem 2:** Necessity follows directly from Theorem 1. For sufficiency, let  $\gamma \in \Gamma$ . Following Theorem 1, we can select  $U^p \in \gamma(\mathbb{R})$ . When constructing the bijection from the uniform distribution on  $(0, 1)$  to the reals, we need to respect map  $\gamma$ . This is always possible since no assumption is made on the type-distribution. ■

**Proof of Theorem 3:** Since the necessity of the axioms is straightforward, we will now prove sufficiency. Consider any menu  $A_j$ . We construct a sequence of open intervals of types,  $\{I_j^0, I_j^1, \dots, I_j^n, \dots\}$ , and a sequence of real functions defined over them,  $\{G_j^0, G_j^1, \dots, G_j^n, \dots\}$ , satisfying the following four properties:

- (1) For every  $n$ ,  $I_j^n \subseteq I_j^{n+1}$ .
- (2) For every  $n$ ,  $G_j^{n+1}$  extends  $G_j^n$ .
- (3) For every  $n$ ,  $G_j^n$  takes values in  $(0, 1)$ , is continuous, and strictly increasing. Moreover, if  $I_j^n$  is bounded from above (respectively, from below), the function  $G_j^n$  must be strictly bounded from above by some value  $k < 1$  (respectively, strictly bounded from below by some value  $k > 0$ ).
- (4) For every  $n$  and every four types  $t_1, t_2, t'_1, t'_2$  in  $I_j^n$ , if  $t_1 + t_2 = t'_1 + t'_2$  then  $\log \frac{G_j^n(t_1)}{1-G_j^n(t_1)} + \log \frac{G_j^n(t_2)}{1-G_j^n(t_2)} = \log \frac{G_j^n(t'_1)}{1-G_j^n(t'_1)} + \log \frac{G_j^n(t'_2)}{1-G_j^n(t'_2)}$ .

The first interval of types,  $I_j^0$ , corresponds to the set of types that have a maximizer in  $A_j \setminus \{\underline{x}_j, \bar{x}_j\}$ .<sup>31</sup> The first function,  $G_j^0$ , corresponds to the function that choice data  $F_j$  induces over these types, i.e., for every  $t \in I_j^0$ ,  $G_j^0(t) = F_j(a_j(\gamma(t)))$ . The function  $G_j^0$  is well-defined given the assumptions made on  $F_j$ . It is obviously strictly increasing and takes values in  $(0, 1)$ . Moreover, if the interval  $I_j^0$  is bounded from above (respectively, from below), there is an interval of types selecting  $\bar{x}_j$  (respectively,  $\underline{x}_j$ ) and property CE implies  $\lim_{a \rightarrow 1} F_j(a) < 1$  (respectively,  $\lim_{a \rightarrow 0} F_j(a) > 0$ ), and hence the boundedness conditions hold for  $G_j^0$ . That is, property (3) is satisfied. Property (4) for  $G_j^0$  follows from CLA by considering  $A_{j'} = A_j$ .

The remaining intervals and functions are now defined recursively. Given collections  $\{I_j^0, I_j^1, \dots, I_j^n\}$  and  $\{G_j^0, G_j^1, \dots, G_j^n\}$  which satisfy all the properties, we define interval  $I_j^{n+1}$  and function  $G_j^{n+1}$  in such a way as to guarantee that collections  $\{I_j^0, I_j^1, \dots, I_j^{n+1}\}$  and  $\{G_j^0, G_j^1, \dots, G_j^{n+1}\}$  also satisfy the properties. The definition of the new interval of types,  $I_j^{n+1}$ , depends on the parity of  $n$ . If  $n$  is an even (respectively, an odd) integer,

<sup>31</sup>Notice that this set of types depends on the assumed type-utility map  $\gamma$ . Since  $\gamma$  is fixed and to simplify the exposition, we will avoid some references to  $\gamma$  in the arguments that follow.

we define interval  $I_j^{n+1}$  as follows: (i) if  $I_j^n$  is not bounded from above (respectively, from below), define  $I_j^{n+1} = I_j^n$  and (ii) if  $I_j^n$  is bounded from above (respectively, from below), define  $I_j^{n+1}$  as the union of the previous interval  $I_j^n$ , the least upper bound (respectively, the greatest lower bound)  $z_j^n$  of interval  $I_j^n$ , and the types  $t$  for which there exists  $t' \in I_j^n$  with  $t = 2z_j^n - t'$ .<sup>32</sup>

We now consider the definition of function  $G_j^{n+1}$ . For every  $t \in I_j^n$ , define  $G_j^{n+1}(t) = G_j^n(t)$ . For the limit type  $z_j^n$ , define  $G_j^{n+1}(z_j^n) = \lim_{s \rightarrow z_j^n} G_j^n(s)$ , where the right-hand or left-hand bound must be considered, depending on the parity. Finally, for any other type  $t$  belonging to  $I_j^{n+1}$ , we know that there exists a unique value  $t' \in I_j^n$  such that  $t = 2z_j^n - t'$ , so we can define  $G_j^{n+1}(t)$  as the unique real value satisfying the equation:

$$\log \frac{G_j^{n+1}(t)}{1 - G_j^{n+1}(t)} = 2 \log \frac{G_j^{n+1}(z_j^n)}{1 - G_j^{n+1}(z_j^n)} - \log \frac{G_j^n(t')}{1 - G_j^n(t')}.$$

It is then evident that the function  $G_j^{n+1}$  is well defined on  $I_j^{n+1}$  and it is straightforward to see that  $I_j^n \subseteq I_j^{n+1}$  and, hence, property (1) holds. Similarly, note that the construction guarantees that the function  $G_j^{n+1}$  extends  $G_j^n$ , and therefore property (2) is satisfied.

We now discuss property (3). Notice that, by the continuity of  $G_j^n$  and the fact that all values belong to  $(0, 1)$ , it is guaranteed that the limit value at  $z_j^n$  is well defined when needed. The continuity of the function  $G_j^{n+1}$  is then a direct consequence of this limit definition at  $z_j^n$ . To appreciate the strictly increasing nature of the new function, consider two types  $t_1 < t_2$ . If both types belong to  $I_j^n$ , we know that  $G_j^{n+1}(t_1) < G_j^{n+1}(t_2)$  must hold because  $G_j^{n+1}$  extends the strictly increasing function  $G_j^n$ . If  $t_1 \in I_j^n$  but  $t_2$  does not, it must be the case that  $n$  is even and there exists  $t'_2 \in I_j^n$  such that  $t_2 = 2z_j^n - t'_2$ . Since  $\log \frac{G_j^n(z_j^n)}{1 - G_j^n(z_j^n)} > \log \frac{G_j^n(t'_2)}{1 - G_j^n(t'_2)}$ , it is  $\log \frac{G_j^{n+1}(t_1)}{1 - G_j^{n+1}(t_1)} = \log \frac{G_j^n(t_1)}{1 - G_j^n(t_1)} < \log \frac{G_j^n(z_j^n)}{1 - G_j^n(z_j^n)} < 2 \log \frac{G_j^n(z_j^n)}{1 - G_j^n(z_j^n)} - \log \frac{G_j^n(t'_2)}{1 - G_j^n(t'_2)} = \log \frac{G_j^{n+1}(t_2)}{1 - G_j^{n+1}(t_2)}$ , as desired. If  $t_1$  is not in  $I_j^n$  but  $t_2$  is, an analogous argument applies in which  $n$  is odd and  $z_j^n$  is the lower bound of  $I_j^n$ . If neither is in  $I_j^n$ , they must both be above or below  $z_j^n$ , depending on the parity. There must exist  $t'_1, t'_2 \in I_j^n$  such that  $t_1 = 2z_j^n - t'_1$  and  $t_2 = 2z_j^n - t'_2$ . It clearly must be that  $t'_1 > t'_2$  and we know that  $G_j^n(t'_1) > G_j^n(t'_2)$ . The definition of  $G_j^{n+1}(t_1)$  and  $G_j^{n+1}(t_2)$  guarantees that the former is strictly smaller than the latter. Hence, we have shown that  $G^{n+1}$  is strictly increasing and, to complete property (3), we need to show

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<sup>32</sup>Intuitively we are extending the original right-bounded (respectively, left-bounded) interval  $I_j^n$  beyond its boundary and adding the boundary point. This step is not needed when there are no corner choices because then the initial interval  $I_j^0$  equals the set of all types,  $\mathbb{R}$ . When choices are observed in only one of the corner alternatives, or, equivalently,  $I_j^0$  is bounded on one side, the logic requires a unique duplication, which already forms the entire real line. If choices are observed in both corner alternatives, or, equivalently, the initial interval is bounded on both sides, we need to duplicate the initial bounded interval an infinite number of times, as the proof indicates.

that this function takes values in  $(0, 1)$  and is bounded as required. We show the case of  $n$  being even, the other case being analogous. If  $I_j^n$  is not bounded from above, the new function replicates the original one and the property holds. If  $I_j^n$  is bounded from above, we know that the value  $G_j^{n+1}(z_j^n)$  must be strictly lower than 1 by virtue of the boundedness condition. For every  $t \in I_j^{n+1}$  with  $t > z_j^n$ , the construction guarantees that  $G_j^{n+1}$  takes values in  $(0, 1)$ . To show boundedness, notice that nothing changes at the lower end of the interval and, since  $G_j^{n+1}$  extends  $G_j^n$ , the property is satisfied. For the upper end of the interval, suppose that  $I_j^{n+1}$  is bounded from above, in which case it must be that  $I_j^n$  is bounded from below (say, with largest lower bound  $k$ ). It then follows that  $\log \frac{G_j^{n+1}(t)}{1-G_j^{n+1}(t)} < 2 \log \frac{G_j^{n+1}(z_j^n)}{1-G_j^{n+1}(z_j^n)} - \log \frac{G_j^n(k)}{1-G_j^n(k)}$ , and hence  $G_j^{n+1}(t)$  must be strictly lower than 1. This completes the proof that  $G_j^{n+1}$  satisfies property (3).

To see that property (4) holds, consider any four types  $t_1, t_2, t'_1, t'_2$  in  $I_j^{n+1}$  such that  $t_1 + t_2 = t'_1 + t'_2$  and assume, without loss of generality, that  $t_1 < t'_1 \leq t'_2 < t_2$ .<sup>33</sup> Again, we show the case of  $n$  even, the other case being analogous. We start by noticing that property (4) holds over the closure of  $I_j^n$ , denoted by  $\bar{I}_j^n$ , thanks to the recursive assumption on  $G_j^n$ , the fact that  $G_j^{n+1}$  extends  $G_j^n$ , and the limit construction at  $z_j^n$ . Hence, we only need to consider cases where not all four types belong to  $\bar{I}_j^n$ :

- Case 1: None of the four types belongs to  $\bar{I}_j^n$ . There must exist  $s_1, s_2, s'_1, s'_2 \in I_j^n$  such that  $t_1 = 2z_j^n - s_1$ ,  $t'_1 = 2z_j^n - s'_1$ ,  $t_2 = 2z_j^n - s_2$  and  $t'_2 = 2z_j^n - s'_2$ . Clearly, it must be that  $s_1 + s_2 = s'_1 + s'_2$  and hence, we know that  $\log \frac{G_j^n(s_1)}{1-G_j^n(s_1)} + \log \frac{G_j^n(s_2)}{1-G_j^n(s_2)} = \log \frac{G_j^n(s'_1)}{1-G_j^n(s'_1)} + \log \frac{G_j^n(s'_2)}{1-G_j^n(s'_2)}$ , which is equivalent to  $\log \frac{G_j^n(s_1)}{1-G_j^n(s_1)} + \log \frac{G_j^n(s_2)}{1-G_j^n(s_2)} + 4G_j^{n+1}(z_j^n) = \log \frac{G_j^n(s'_1)}{1-G_j^n(s'_1)} + \log \frac{G_j^n(s'_2)}{1-G_j^n(s'_2)} + 4G_j^{n+1}(z_j^n)$ , which implies  $\log \frac{G_j^n(t_1)}{1-G_j^n(t_1)} + \log \frac{G_j^n(t_2)}{1-G_j^n(t_2)} = \log \frac{G_j^n(t'_1)}{1-G_j^n(t'_1)} + \log \frac{G_j^n(t'_2)}{1-G_j^n(t'_2)}$ , as desired.
- Case 2:  $t_1 \in \bar{I}_j^n$ . There must exist  $s_2, s'_1, s'_2 \in I_j^n$  such that  $t'_1 = 2z_j^n - s'_1$ ,  $t_2 = 2z_j^n - s_2$  and  $t'_2 = 2z_j^n - s'_2$ . It must be that  $t_1 + s'_1 + s'_2 = s_2 + 2z_j^n$ . Define  $\hat{t} = s_2 + z_j^n - t_1$ , which belongs to  $I_j^n$ . Given that  $t_1 + \hat{t} = s_2 + z_j^n$ , property (4) holds over these four types. Now, notice that it must also be that  $s'_1 + s'_2 = \hat{t} + z_j^n$  and hence property (4) holds over these four types. We can combine the two expressions to verify that property (4) holds over  $t_1, t_2, t'_1$  and  $t'_2$ , as desired.
- Case 3:  $t_1, t'_1 \in \bar{I}_j^n$ . There must exist  $s_2, s'_2 \in I_j^n$  such that  $t_2 = 2z_j^n - s_2$  and  $t'_2 = 2z_j^n - s'_2$ . It must be that  $t_1 + s'_2 = t'_1 + s_2$  and hence, we know that  $\log \frac{G_j^n(t_1)}{1-G_j^n(t_1)} + \log \frac{G_j^n(s'_2)}{1-G_j^n(s'_2)} = \log \frac{G_j^n(t'_1)}{1-G_j^n(t'_1)} + \log \frac{G_j^n(s_2)}{1-G_j^n(s_2)}$ , which implies  $\log \frac{G_j^n(t_1)}{1-G_j^n(t_1)} +$

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<sup>33</sup>Notice that if the types were equal across the two pairs, the property would be trivially satisfied.

$\log \frac{G_j^n(s'_2)}{1-G_j^n(s'_2)} + 2G_j^{n+1}(z_j^n) = \log \frac{G_j^n(t'_1)}{1-G_j^n(t'_1)} + \log \frac{G_j^n(s_2)}{1-G_j^n(s_2)} + 2G_j^{n+1}(z_j^n)$ , which implies  $\log \frac{G_j^n(t_1)}{1-G_j^n(t_1)} + \log \frac{G_j^n(t_2)}{1-G_j^n(t_2)} = \log \frac{G_j^n(t'_1)}{1-G_j^n(t'_1)} + \log \frac{G_j^n(t'_2)}{1-G_j^n(t'_2)}$ , as desired.

- Case 4:  $t_1, t'_1, t'_2 \in \bar{I}_j^n$ . There must exist  $s_2 \in I_j^n$  such that  $t_2 = 2z_j^n - s_2$ . It must be that  $t_1 + 2z_j^n = t'_1 + t'_2 + s_2$ . Define  $\hat{t} = t_1 + z_j^n - t'_1$ , which belongs to  $I_j^n$ . Given that  $t'_1 + \hat{t} = t_1 + z_j^n$ , property (4) holds over these four types. Now, notice that it must also be that  $\hat{t} + z_j^n = t'_2 + s_2$  and hence property (4) holds over these four types. We can combine the two expressions to verify that property (4) holds over  $t_1, t_2, t'_1$  and  $t'_2$ , as desired.

This completes the proof that the collections  $\{I_j^0, I_j^1, \dots, I_j^{n+1}\}$  and  $\{G_j^0, G_j^1, \dots, G_j^{n+1}\}$  satisfy all the properties. The limit interval of the sequence  $\{I_j^0, I_j^1, \dots, I_j^n, \dots\}$  is the entire set of reals. The limit function of the sequence  $\{G_j^0, G_j^1, \dots, G_j^n, \dots\}$ , which we denote by  $G_j$ , must be a continuous, strictly increasing CDF over the reals. Moreover, it extends  $G_j^0$  and must also satisfy property (4) above.

Consider the median type of distribution  $G_j$ , i.e., the type  $\tau_j$  such that  $G_j(\tau_j) = .5$ . Define the function  $H_j$  over the reals as follows:

$$H_j(w) = G_j(\tau_j + w).$$

We claim that  $H_j$  is a continuous, strictly increasing CDF over the reals, and is symmetric with respect to the origin. We need to show symmetry. For this, consider  $t_1 = \tau_j - w$ ,  $t_2 = \tau_j + w$  and  $t'_1 = t'_2 = \tau_j$ . Then, since  $t_1 + t_2 = t'_1 + t'_2$ , we know that  $\log \frac{G_j(t_1)}{1-G_j(t_1)} + \log \frac{G_j(t_2)}{1-G_j(t_2)} = \log \frac{G_j(t'_1)}{1-G_j(t'_1)} + \log \frac{G_j(t'_2)}{1-G_j(t'_2)} = 0 + 0 = 0$ . Hence, it must be that  $\log \frac{G_j(t_1)}{1-G_j(t_1)} = \log \frac{1-G_j(t_2)}{G_j(t_2)}$  and  $G_j(t_1) = 1 - G_j(t_2)$  follows. As a result,  $H_j(-w) = G_j(t_1) = 1 - G_j(t_2) = 1 - H_j(w)$ , and the symmetry of  $H_j$  has been proved.

Consider now the following function defined over the positive reals:

$$O_j(w) = \frac{1 - H_j(w)}{H_j(w)}.$$

Since  $H_j$  is a continuous, strictly increasing CDF over the reals with  $H_j(0) = .5$ ,  $1 - O_j(w)$  must be a continuous, strictly increasing CDF over the positive reals with no strictly positive mass at zero. Moreover, given that  $G_j$  satisfies property (4) above, the definition of  $H_j$  and  $O_j$  guarantees that  $O_j(w)O_j(z) = O_j(w+z)$  must hold for every pair of positive real values  $w$  and  $z$ . One can then reproduce the standard argument dating back to Cauchy (1821), which is described in Galambos and Kotz (1978; Theorem 1.3.1), to guarantee that  $O_j$  must be an exponential distribution, with no strictly positive mass at the origin.<sup>34</sup> That is, there exists  $\sigma_j \in \mathbb{R}_{++}$  such that

$$1 - O_j(w) = 1 - \frac{1 - H_j(w)}{H_j(w)} = 1 - e^{-w/\sigma_j},$$

<sup>34</sup>The property is satisfied by exponential distributions with and without strictly positive mass at zero. Since we know that  $O$  has no strictly positive mass at zero, it must be one of the latter.

and hence, for every  $w \geq 0$ , it is true that  $H_j(w) = \frac{1}{1+e^{-w/\sigma_j}}$ . Moreover, given the symmetry of  $H_j$  with respect to the origin, for every  $w < 0$ , it must also be true that  $H_j(w) = 1 - H_j(-w) = 1 - \frac{1}{1+e^{w/\sigma_j}} = \frac{1}{1+e^{-w/\sigma_j}}$ . That is,  $H_j$  is a logistic distribution with location parameter equal to zero and scale parameter  $\sigma_j$ , and  $G_j$  is ordered logistic with location parameter  $\tau_j$  and scale parameter  $\sigma_j$ . Since  $G_j$  extends  $G_j^0$ , all choices in menu  $A_j$  are explained by this distribution.

Consider now two menus  $A_j$  and  $A_{j'}$ . By our richness assumption, there exists a sequence of menus  $A_{j_0} = A_j, A_{j_1}, \dots, A_{j_k}, \dots, A_{j_K} = A_{j'}$  such that, for every  $k \in \{0, \dots, K-1\}$ ,  $I_{j_k}^0 \cap I_{j_{k+1}}^0 \neq \emptyset$ . Consider  $t \in I_{j_k}^0 \cap I_{j_{k+1}}^0$  and take  $t_1 = t_2 = t'_1 = t'_2 = t$ . Using the ordered-logit structure of  $G_{j_k}$  and  $G_{j_{k+1}}$ , it follows that they must both have a common location parameter  $\tau$  and a common scale parameter  $\sigma$ . The recursive application of this argument shows that  $G_j$  and  $G_{j'}$  must have the same common parameters  $\tau$  and  $\sigma$ , which concludes the proof. ■

**Proof of Corollary 2:** The first part follows immediately from Theorem 1 in Kubler, Selden and Wei (2014) and from our Theorem 1. The second part follows from our Theorem 2 after noticing that  $\frac{\log \phi}{\log r}$  represents the negative of the curvature of the CRRA function maximizing at the consumption ratio  $r$ , with strictly negative values representing risk-aversion and null values corresponding to any other risk attitude. ■

**Proof of Corollary 4:** The first part follows immediately from classical deterministic results on consumption theory. Notice that utilities that are strictly monotone in the first component always select the corner  $x^2 = 0$ , and generate choices that are consistent with WARP. Also, utilities that are strictly increasing in both components and strictly convex must also lead to choices consistent with WARP. Moreover, satisfaction of WARP allows to rationalize choices with such utility functions (see, e.g. Rose (1958) and Matzkin and Richter (1991)). Then, the application of our Theorem 1 concludes the proof. For the second part, notice that when  $t \leq 0$ , all choices are such that  $x^2 = 0$ . When  $t > 0$ , the first-order condition analysis for a given budget set allows to obtain  $t = \pi_j r_j(x)^{1-\alpha}$ . By equalizing this value for two different budget sets  $A_j, A_{j'}$ , one can then see that  $\alpha = \frac{\log(\pi_j/\pi_{j'})}{\log(r_j(x)/r_{j'}(x'))} + 1$ , which must then be constant across pairs of menus. If deterministic data has this constancy with interior solutions, one can define  $\alpha$  to be such constant, and then define  $t$  from one menu, using  $\alpha$ . For corner solutions, one can set  $t \leq 0$  and choose freely  $\alpha$ . The utility function defined in such ways rationalizes the observations. To apply our Theorem 1, notice that we want the utility function to use a constant  $\alpha$  across quantiles, which is guaranteed by our assumption. ■

## APPENDIX B. DISCRETE CHOICE

We present here some simple discrete versions of our results, serving several purposes. First, they illustrate that our analysis is not confined to continuous settings.

Second, they may be useful for practitioners working with discrete choice environments. Third, they help provide a more comprehensive understanding of the theoretical results developed in this paper in relation to existing literature.

Let  $(X, <)$  be a finite, partially ordered set of alternatives. Denote by  $\mathcal{X}$  the set of all subsets of  $X$  containing at least two alternatives, and by  $\mathcal{D} \subseteq \mathcal{X}$  an arbitrary domain consisting of fully ordered menus, i.e., the restriction of  $<$  to any  $A \in \mathcal{D}$  is complete. An ordered menu  $A_j$  can be written as  $A_j = \{1_j, 2_j, \dots, i_j, \dots, I_j + 1\}$ , where  $1_j < 2_j < \dots < i_j < \dots < I_j + 1$ .

Choice data is described by a stochastic choice function  $\rho$  defined on  $\mathcal{D}$ . Specifically,  $\rho(x, A) \in [0, 1]$ , with  $x \in A$  and  $A \in \mathcal{D}$ , represents the probability of choosing alternative  $x$  from menu  $A$ , with the constraint  $\sum_{x \in A} \rho(x, A) = 1$ . We denote cumulative choice data by  $\bar{\rho}$ , defined as  $\bar{\rho}(i_j, A_j) = \sum_{k \leq i_j} \rho(k, A_j)$ .

The definition of ORUM-rationalizability in this discrete setting follows directly from the one in the main text, with  $F$  replaced by  $\bar{\rho}$ . Given the finite nature of this setting, the image of  $\gamma$  can be interpreted as a finite ordered collection of linear orders, with  $G$  being the CDF of a finite probability distribution  $g$ .

**B.1. Nonparametric case.** To present a version of Theorem 1 in this discrete setting, we define the  $p$ -quantile choice function  $c^p$ . Formally,  $c^p(A)$  is the first alternative in  $A$  that reaches cumulative probability mass  $p$ , i.e.,  $\bar{\rho}(c^p(A) - 1, A) < p \leq \bar{\rho}(c^p(A), A)$ .<sup>35</sup> A quantile choice function  $c^p$  is said to be rationalizable if there exists a linear order  $P$  on  $X$  such that, for every  $A \in \mathcal{D}$ ,  $c^p(A) = m(A, P)$ , i.e.,  $c^p(A)$  is the maximal element in  $A$  according to  $P$ . It is well known that, in arbitrary and finite domains, deterministic rationalizability is equivalent to the acyclicity notion used in the main text.

**Proposition 1.**  *$\rho$  is ORUM-rationalizable if and only if  $\rho$  satisfies Quantile Acyclicity.*

**Proof of Proposition 1:** Suppose first that  $\rho$  is ORUM-rationalizable by means of  $(\gamma, g)$ . Given the finiteness of the setting,  $(\gamma, g)$  induces a finite, ordered collection of linear orders  $\{P_t\}_{t=1}^T$ , with increasing maximizers in every menu, and a finite probability distribution over  $\{1, \dots, T\}$ , denoted by  $g$  for simplicity. Consider any  $p \in (0, 1)$ , and let  $t(p)$  be such that  $\sum_{t < t(p)} g(t) < p \leq \sum_{t \leq t(p)} g(t)$ . Given the increasing nature of the maximizers of  $\{P_t\}_{t=1}^T$ , it follows that, for every  $A \in \mathcal{D}$ , the quantile choices  $\{c^p(A)\}_{A \in \mathcal{D}}$  coincide with the maximizers of  $P_{t(p)}$ . Hence, these choices must be rationalizable and, by classical results, must satisfy Acyclicity.

Conversely, suppose that  $c^p$  satisfies Acyclicity for every  $p \in (0, 1)$ . These choices can be rationalized by a linear preference, denoted  $P_p$ . Since the setting is finite, there are only finitely many distinct linear preferences in the family  $\{P_p\}_{p \in (0, 1)}$ , denoted  $\{P_t\}_{t=1}^T$ . By discarding preferences that generate identical maximizers in the domain  $\mathcal{D}$ , we

<sup>35</sup>When  $c^p(A) = 1_j$ , we set  $\bar{\rho}(c^p(A) - 1, A) = 0$ . Similarly,  $i_j - 1$  is denoted as  $(i - 1)_j$ . In the proof below, when  $t(p) = 1$ , we write  $\sum_{t < t(p)} g(t) = 0$ .

can assume, without loss of generality, that distinct preferences in this collection yield different choice functions. Furthermore, given the quantile definition, each of these preferences must correspond to an interval of  $p$ -values. Thus, we can assume that the family  $\{P_t\}_{t=1}^T$  is ordered, i.e.,  $P_t$  corresponds to lower  $p$ -values than  $P_{t'}$  whenever  $t < t'$ . We can then assign to each preference  $P_t$  the Lebesgue measure of the corresponding interval of  $p$ -values, denoted  $g(t)$ . This pair  $(\{P_t\}_{t=1}^T, g)$  rationalizes the data, so  $\rho$  is ORUM-rationalizable. ■

Proposition 1, a discrete version of Theorem 1 in the main text, clarifies the difference with the SCRUM model of Apesteguia, Ballester, and Lu (2017). Essentially, ORUM-rationalization is a more general model that allows for  $X$  to be partially ordered and the consideration of an arbitrary domain of menus.

Proposition 2 below shows that when  $X$  is fully ordered and choices are observed from every subset of  $X$ , ORUM-rationalization reduces to SCRUM-rationalization. To do so, we use the discrete version of the Weak Axiom of Revealed Preference (D-WARP) at the quantile level, where the direct revelations of the quantile choice function  $c^p$  must contain no cycles, and property  $\alpha$ , where  $c^p(A) \in B \subseteq A$  implies  $c^p(B) = c^p(A)$ . We say that  $\rho$  satisfies Quantile D-WARP (respectively, Quantile  $\alpha$ ) if every  $c^p$  satisfies WARP (respectively, property  $\alpha$ ). Consider also the two properties characterizing SCRUMs in Apesteguia, Ballester, and Lu (2017): Regularity and Centrality.<sup>36</sup>

**Proposition 2.** *Let  $\mathcal{D} = \mathcal{X}$ .  $\rho$  is ORUM-rationalizable (or SCRUM-rationalizable) if and only if  $\rho$  satisfies Quantile D-WARP, if and only if  $\rho$  satisfies Quantile  $\alpha$ , if and only if  $\rho$  satisfies Regularity and Centrality.*

**Proof of Proposition 2:** Under  $\mathcal{D} = \mathcal{X}$ , classical results ensure that  $c^p$  satisfies Acyclicity if and only if it satisfies D-WARP, if and only if it satisfies property  $\alpha$ , allowing us to apply Proposition 1. To prove the final statement, note that since  $\mathcal{D} = \mathcal{X}$ , all binary sets are part of the domain, and hence  $X$  must be completely ordered. It follows that ORUM-rationalization is equivalent to SCRUM-rationalization (a probability distribution over a single-crossing collection of preferences). The characterization result in Apesteguia, Ballester, and Lu (2017) completes the proof. ■

Thanks to Proposition 2, one can also see better how the properties of nonparametric ORUMs relate to the existing properties of SCRUMs. To illustrate, we show how the main property of SCRUMs, Centrality, is implied by Quantile Acyclicity in this restricted setting. Consider three alternatives,  $x < y < z$ , and suppose that the probability of choosing  $y$  from menu  $\{x, y, z\}$  is strictly positive, i.e.,  $\rho(y, \{x, y, z\}) > 0$ . Suppose, by way of contradiction, that  $\rho(z, \{x, y, z\}) \neq \rho(z, \{y, z\})$ . If  $\rho(z, \{x, y, z\}) > \rho(z, \{y, z\})$ , it must be  $\rho(x, \{x, y, z\}) + \rho(y, \{x, y, z\}) < \rho(y, \{y, z\})$  and the quantile definition together with the ordered nature of choices guarantee the existence of a value  $p$  such that  $c^p(\{x, y, z\}) = z$ , but  $c^p(\{y, z\}) = y$ . If  $\rho(z, \{x, y, z\}) < \rho(z, \{y, z\})$ , and

<sup>36</sup>Centrality:  $x < y < z$  or  $z < y < x$  and  $\rho(y, \{x, y, z\}) > 0$  imply  $\rho(z, \{x, y, z\}) = \rho(z, \{y, z\})$ .

given that  $\rho(y, \{x, y, z\}) > 0$ , there must exist some  $p$  such that  $c^p(\{x, y, z\}) = y$ , but  $c^p(\{y, z\}) = z$ . In both cases, there is a contradiction with the fact that  $c^p$  satisfies the classical property  $\alpha$  and, consequently, a violation of the acyclicity of  $c^p$ .

**B.2. Semi-nonparametric case.** We now present a discrete choice version of the semi-nonparametric results in the main text, which allows us to relate the present results to Apesteguia and Ballester (2023). To do so, fix a parametrization  $\gamma$  that generates ordered choices. That is, in any given menu  $A_j \in \mathcal{D}$ , types  $(-\infty, t_j^1)$  uniquely select  $1_j$ , types  $(t_j^1, t_j^2)$  uniquely select  $2_j$ , and so on and so forth, with types  $(t_j^{I_j}, \infty)$  uniquely selecting  $I_j + 1$ .<sup>37</sup> Let the utility functions in  $\gamma$  be characterized by a certain deterministic property, that we call  $\gamma(\mathbb{R})$ -Rationalizability and define Quantile  $\gamma(\mathbb{R})$ -Rationalizability accordingly.

The Monotonicity property of Apesteguia and Ballester (2023) reads as follows: for every  $B \subseteq A \in \mathcal{D}$  and  $B' \subseteq A' \in \mathcal{D}$ , if the set of types maximizing in  $B$  within  $A$  is included in the set of types maximizing in  $B'$  within  $A'$ , then  $\sum_{x \in B} \rho(x, A) \leq \sum_{x \in B'} \rho(x, A')$ .

**Proposition 3.** *Let  $\gamma$  be any type-utility map.  $\rho$  is ORUM-rationalizable with type-utility map  $\gamma$  if and only if every  $\rho$  satisfies Quantile  $\gamma(\mathbb{R})$ -Rationalizability, if and only if  $\rho$  satisfies Monotonicity.*

**Proof of Proposition 3** The result follows immediately from Proposition 1 and the characterization result of Apesteguia and Ballester (2023). ■

Proposition 3 shows that the model of Apesteguia and Ballester (2023) is equivalent to a discrete choice version of the semi-nonparametric model described in Theorem 2. The connection between the quantile rationalizability approach of the present paper, and the property of Monotonicity can be presented as follows. Suppose that  $\rho$  satisfies quantile rationalizability and assume, by way of contradiction, that Monotonicity fails. When Monotonicity fails, it can be shown that a cumulative violation of Monotonicity must exist. Namely, there exist  $(i_j, A_j)$  and  $(i'_{j'}, A_{j'})$  such that the types maximizing below  $i_j$  in  $A_j$  contain the types maximizing below  $i'_{j'}$  in  $A_{j'}$ , i.e.,  $t_j^i \geq t_{j'}^{i'}$ , but we observe  $\bar{\rho}(i_j, A_j) < \bar{\rho}(i'_{j'}, A_{j'})$ . That is, every preference with property  $\gamma(\mathbb{R})$ -Rationalizability maximizing below  $i'_{j'}$  in  $A_{j'}$  must also maximize below  $i_j$  in  $A_j$ . Hence, given the inequality in cumulative choice probabilities the quantile choice function  $c^{\rho(i'_{j'}, A_{j'})}$  violates property  $\gamma(\mathbb{R})$ -Rationalizability, a contradiction.

**B.3. Parametric case.** We now consider the parametric version of ordered logit in a discrete setting. To do so, fix a map  $\gamma$  and assume the logistic distribution. In the

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<sup>37</sup>Since in the continuous setting types have no mass, the choice at threshold types is irrelevant and ties can be avoided. Moreover, we simplify notation and discard from a menu alternatives that are not maximal for any type.

ordered-logit model the cumulative choice probability of alternatives  $\{1_j, 2_j, \dots, i_j\}$  in decision problem  $A_j$  is determined by the threshold type  $t_j^i$  as

$$G^{(\tau, \sigma)}(t_j^i) = \frac{1}{1 + e^{-(t_j^i - \tau)/\sigma}}.$$

The ordered-logit rationalization of choice data requires, therefore, that  $G^{(\tau, \sigma)}(t_j^i) = \bar{\rho}(i_j, A_j)$ .

We assume Positivity that, given the ordered nature of the setting, is equivalent to  $0 < \bar{\rho}(1_j, A_j) < \dots < \bar{\rho}(i_j, A_j) < \dots < \bar{\rho}(I_j, A_j) < 1 = \bar{\rho}(I_j + 1, A_j)$  for every  $A_j \in \mathcal{D}$ . In addition, we make the following technical assumption: there exist threshold types  $t_{j_1}^{i_1}, t_{j_2}^{i_2}, t_{j_3}^{i_3}, t_{j_4}^{i_4}$ , and  $t_{j^*}^{i^*}$  such that  $\bar{\rho}(i_1, A_{j_1}) < \bar{\rho}(i^*, A_{j^*}) = .5 < \bar{\rho}(i_2, A_{j_2})$ , and the value  $\frac{t_{j_3}^{i_3} - t_{j^*}^{i^*}}{t_{j_4}^{i_4} - t_{j^*}^{i^*}}$  is an irrational number.

We introduce a version of CLA that characterizes ordered-logistic choice in discrete settings. Consider any two equally-sized collections of threshold types. The property states that the sum of the first collection is larger than that of the second if and only if the corresponding sum of cumulative log-odds in the former is larger than that in the latter.<sup>38</sup>

**Discrete Cumulative Log-Odds Additivity (D-CLA).** For every positive integer  $M$ , and for every two collections of threshold types  $\{t_{j_1}^{i_1}, \dots, t_{j_m}^{i_m}, \dots, t_{j_M}^{i_M}\}$  and  $\{t_{j'_1}^{i'_1}, \dots, t_{j'_m}^{i'_m}, \dots, t_{j'_M}^{i'_M}\}$ ,  $\sum_{m=1}^M t_{j_m}^{i_m} \geq \sum_{m=1}^M t_{j'_m}^{i'_m}$  if and only if  $\sum_{m=1}^M \ell_{j_m}(i_m) \geq \sum_{m=1}^M \ell_{j'_m}(i'_m)$ .

**Proposition 4.** *Let  $\gamma$  be any type-utility map.  $\rho$  is ORUM-rationalizable with type-utility map  $\gamma$  and a type distribution in  $\mathcal{G}^L$  if and only if  $\rho$  satisfies D-CLA.*

**Proof of Proposition 4:** In a discrete setting, it is immediate to see that any choice data that is rationalized by an ordered-logit model must satisfy D-CLA. We then need to prove the sufficiency part of the result. For this, we start by constructing a function over the reals. By assumption, there is a threshold type  $t_{j^*}^{i^*}$  such that  $\bar{\rho}(i^*, A_{j^*}) = .5$ . Consider then the subsets of real numbers

$$\begin{aligned} \mathcal{T} &= \{x : x = t_j^i - t_{j^*}^{i^*} \text{ for some } A_j \in \mathcal{D} \text{ and } i_j < I_j + 1\}, \\ \mathcal{T}^{IC} &= \{x : x \text{ is an integer combination of elements in } \mathcal{T}\}. \end{aligned}$$

It is immediate to see that  $\mathcal{T}^{IC}$  is a subgroup of the reals, and given our technical assumption on the existence of threshold types producing a ratio that is irrational, well-known results guarantee that  $\mathcal{T}^{IC}$  must be dense in the reals.<sup>39</sup> We can then find, for every  $x \in \mathbb{R}$ , a sequence of elements  $(x_1, x_2, \dots, x_k, \dots)$  in  $\mathcal{T}^{IC}$  such that  $x_k \rightarrow x$ . Each of the elements  $x_k$  in this sequence is an integer combination of elements

<sup>38</sup>The cumulative log-odds in the discrete setting follow from the definition used in the continuous setting, with  $F$  replaced by  $\bar{\rho}$ . Notice that the continuous setting helps in the presentation of the property by focusing on pairs of types.

<sup>39</sup>See, e.g., Theorem 1.6 in Salzman, Grundhöfer, Hähl and Löwen (2007).

in  $\mathcal{T}$  and hence, we can find collections of threshold types  $\{t_{j_1}^{i_1}, \dots, t_{j_v}^{i_v}, \dots, t_{j_V}^{i_V}\}$  and  $\{s_{j_1}^{i_1}, \dots, s_{j_w}^{i_w}, \dots, s_{j_W}^{i_W}\}$  such that  $x_k = \sum_{v=1}^V n_v(t_{j_v}^{i_v} - t_{j^*}^{i^*}) - \sum_{w=1}^W n_w(s_{j_w}^{i_w} - t_{j^*}^{i^*})$ , where all  $n_v$  and  $n_w$  are strictly positive integers. Consider the real value  $H_k(x)$  that solves the equality  $\log \frac{H_k(x)}{1-H_k(x)} = \sum_{v=1}^V n_v \ell_{j_v}(i_v) - \sum_{w=1}^W n_w \ell_{j_w}(i_w)$ . Denoting by  $H(x)$  the limit of the sequence of values formed by  $H_k(x)$ , we have constructed a function  $H$  over the reals.

First of all, notice that for any given threshold type  $t$ , there may be several decision problems which have this value  $t$  as a threshold type. D-CLA guarantees that the cumulative choice probability is the same in both decision problems, and hence the function  $H$  is well defined.<sup>40</sup> We now prove that  $H$  is increasing. Let  $x < x'$ . We know that there exist sequences of elements  $(x_1, x_2, \dots, x_k, \dots)$  and  $(x'_1, x'_2, \dots, x'_k, \dots)$  in  $\mathcal{T}^{IC}$  such that  $x_k \rightarrow x$  and  $x'_k \rightarrow x'$ . Since  $x < x'$ , there exists  $K$  such that  $x_k < x'_k$  for every  $k \geq K$ . Let  $k \geq K$ , and consider the integer representations of  $x_k$  given by  $\{n_v, t_{j_v}^{i_v}\}_{v=1}^V$  and  $\{n_w, s_{j_w}^{i_w}\}_{w=1}^W$  and of  $x'_k$  given by  $\{n'_v, t_{j'_v}^{i'_v}\}_{v=1}^{V'}$  and  $\{n'_w, s_{j'_w}^{i'_w}\}_{w=1}^{W'}$ . Consider the two positive integer values  $\sum_{v=1}^V n_v + \sum_{w=1}^W n'_w$  and  $\sum_{v=1}^{V'} n'_v + \sum_{w=1}^W n_w$ , one of which must be larger than the other. Consider w.l.o.g, that the former is the larger and, on this basis, construct the following two collections of threshold types. In collection one, we perform  $n_v$  repetitions, from  $v = 1$  to  $V$ , of the threshold type  $t_{j_v}^{i_v}$ , and  $n'_w$  repetitions, from  $w = 1$  to  $W'$ , of the threshold type  $s_{j'_w}^{i'_w}$ . In the second collection, we perform  $n'_v$  repetitions, from  $v = 1$  to  $V'$ , of the threshold type  $t_{j'_v}^{i'_v}$ ;  $n_w$  repetitions, from  $w = 1$  to  $W$ , of the threshold types  $s_{j_w}^{i_w}$ ; and, finally,  $\sum_{v=1}^V n_v + \sum_{w=1}^{W'} n'_w - \sum_{v=1}^{V'} n'_v - \sum_{w=1}^W n_w$  repetitions of the threshold type  $t_{j^*}^{i^*}$ . By construction, these two collections have the same number of components, all of which are threshold types. Moreover, given that  $x_k < x'_k$ , the sum of types is strictly smaller in the former collection than in the latter, and the application of D-CLA guarantees that the sum of cumulative log-odds is strictly smaller in the former collection than in the latter. The limit definition of  $H$  guarantees that  $H(x) \leq H(x')$ . Similarly, it is immediate to see that  $H$  is continuous and, given the definition of  $t_{j^*}^{i^*}$ , it is also obvious that  $H(0) = .5$ . Moreover, by assumption, there are threshold types  $t_{j_1}^{i_1}$  and  $t_{j_2}^{i_2}$  such that  $\bar{\rho}(i_1, A_{j_1}) < \bar{\rho}(i^*, A_{j^*}) = .5 < \bar{\rho}(i_2, A_{j_2})$  and hence by taking into account the sequences of real numbers given by  $\{t_{j_1}^{i_1} - t_{j^*}^{i^*}, 2(t_{j_1}^{i_1} - t_{j^*}^{i^*}), \dots, k(t_{j_1}^{i_1} - t_{j^*}^{i^*}), \dots\}$  and  $\{t_{j_2}^{i_2} - t_{j^*}^{i^*}, 2(t_{j_2}^{i_2} - t_{j^*}^{i^*}), \dots, k(t_{j_2}^{i_2} - t_{j^*}^{i^*}), \dots\}$ , it is obvious that  $H$  approaches 0 (respectively 1) when considering real values approaching  $-\infty$  (respectively,  $\infty$ ). It is also immediate to see that  $H$  satisfies property 4 as described in the proof of Theorem 3.

Consider now the following function defined over the positive reals:

$$O(x) = \frac{1 - H(x)}{H(x)}.$$

<sup>40</sup>Indeed, the same idea applies to the extension of  $H$  to any real number, by using limits of integer combinations of threshold types. This argument is omitted below.

From the properties of  $H$ , it is immediate that  $1 - O(x)$  must be a continuous CDF over the positive reals and that  $O(x)O(z) = O(x + z)$  must hold for every pair of positive real values  $x$  and  $z$ . The same arguments used in the proof of Theorem 3 can be used to show the logistic nature of  $H$ , as desired. ■

Proposition 4 is the discrete version of Theorem 3. Note that the strategies followed in the corresponding proofs are rather different. In the case of continuous choice, choice data from a decision problem provide information on an interval of types that must be extended to the real line in a way that satisfies the additive requirements of CLA. Then, we use the intersection of the intervals of types across menus and the CLA property to guarantee that all these extensions follow the same logistic distribution. However, each discrete choice problem provides information only over a finite number of thresholds. Given the sparsity of these thresholds, we need to consider all of them together, and expand the information to the real line. We do this by using integer combinations of these thresholds, which, from classic results, are known to form a dense subset of the reals. In order to implement this strategy, we need a stronger additivity property, D-CLA, which operates not only over pairs of types but over two equally-sized collections of types. Once the extension to the reals is done, we can complete the proof using arguments from the proof of Theorem 3.<sup>41</sup>

## APPENDIX C. MIXED ORDERED LOGIT

It is well-known that RUMs can be approximated by mixed-logit models (see, McFadden and Train (2000)). We show here that the same kind of approximation can be established in the ORUM framework. Given Corollary 1 we only need to discuss the case in which the type-utility map  $\gamma$  is fixed. Denote by  $\mathcal{G}^{ML}$  the set of all possible mixed-logistic distributions, i.e., the convex hull of  $\mathcal{G}^L$ .

**Theorem 4.** *Let  $\gamma$  be any type-utility map. If  $F$  is ORUM-rationalizable with type-utility map  $\gamma$ , there is a sequence  $\{G^1, \dots, G^n, \dots\}$  with  $G^n \in \mathcal{G}^{ML}$ , such that  $F^n \rightarrow F$ , where  $F^n$  is ORUM-rationalizable with type-utility map  $\gamma$  and type distribution  $G^n$ .*

**Proof of Theorem 4:** Consider any  $F$  that is ORUM-rationalizable with type-utility map  $\gamma$ . We first show that for any utility function  $U$  in the image of  $\gamma$ , i.e.,  $\gamma(t) = U$  for some  $t$ , there exists a sequence of ordered logits producing choice distributions that converge to the deterministic choice function generated by  $U$ . Consider a sequence of ordered logits, all of them using map  $\gamma$ , in which the  $n$ -th element of the sequence has mean  $\tau_n = t$  and standard deviation  $\frac{1}{n}$ . For every menu  $A_j$ , consider the element  $x_j$  such that  $a_j(x_j) = a_j(U)$ . It is immediate that for every  $\epsilon > 0$ , we can find a neighborhood of  $a_j(x_j)$  such that the cumulative choice probability in that neighborhood approaches

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<sup>41</sup>As in the continuous case, the discrete setting also has a unique pair of parameters  $(\tau, \sigma)$  rationalizing the data.

1 whenever  $n \rightarrow \infty$ . Given finiteness of the domain of menus, this concludes the proof of the claim.

We now prove the main statement. Let  $G$  be the type distribution rationalizing  $F$ . For every  $U$  in the image of  $\gamma$ , consider the sequence of real-valued intervals  $\{I_n\}$ , where  $I_n = [-n, n]$ . Consider a sequence of finite subsets of real numbers  $K_n$ , where  $K_n$  is formed by those elements resulting from a partitioning of  $I_n$  into  $n^2$  equal-size intervals. This sequence has the property of both expanding its support and becoming finer when  $n$  grows. For every value of  $n$ , consider the following mixed ordered logit: for every  $t \in K_n$ , use the ordered logit such that  $\tau = t$  and  $\sigma = \frac{1}{n}$ . When  $n$  grows large, we know that this ordered logit approximates the choice frequencies of utility  $\gamma(t)$ . Denote by  $(t', t'')$  the interval formed by the elements in  $K_n$  right before and right after  $t$ . Given continuity, this ordered logit approximates well the choice frequencies generated by the truncated distribution over types  $(t', t'')$ . To each sub-interval, assign the cumulative mass assigned that corresponds to  $\gamma$  and  $G$ . Given finiteness of the domain of menus, the mixed ordered logit approximates well the choice frequencies predicted by the truncation of the ORUM to interval  $I_n$ . The result follows. ■

#### APPENDIX D. EMPIRICAL ILLUSTRATION

In this section, we revisit the theoretical applications developed in Section 6 but from a more applied perspective. Our aim is threefold: to illustrate the types of data consistent with our theoretical models, to show how such data can be treated and interpreted in light of the theory, and to offer some initial ideas for testing. In doing the latter, we emphasize that we do not introduce any new econometric machinery, as this lies outside the scope of the paper. The simplicity of our theoretical framework, type-utility maps, and types distributions makes it possible to apply standard techniques. In fact, most of the methods rely on well-known results from the literature, including classical algorithms, tests for the equality of empirical distributions, or conventional maximum-likelihood estimation with censored data. This simplicity is, in our view, a strength of the theoretical analysis presented in the paper. It shows that familiar tools can take applied researchers a long way in testing the empirical content of ORUMs. Naturally, future econometric work should aim to develop asymptotically valid and more efficient tests tailored to the ORUM framework, especially in its nonparametric version.

In what follows, we describe our data-generating process, and for each characterization result (nonparametric, semi-parametric, and parametric) and each application (risk and altruism), we discuss the use of data in the empirical strategy, some preliminary tests, and the results of our analysis. We conclude with a discussion of some of the challenges that future developments of econometric tools tailored to the ORUM framework may face.

**D.1. Data.** For each application, we generate  $J = 20$  menus of the type described in Sections 6.1 and 6.2. Prices are randomly selected from the uniform distribution in the range  $[1/100, 1/10]$ . In the risk application, we further draw the probabilities of state 1,  $q_j^1$ , randomly from the uniform distribution  $[1/5, 4/5]$ , and relabel the menus such that state 1 pays more in expectation, i.e.,  $\phi_j^1 \equiv \frac{\pi_j^1}{q_j^1} < \frac{\pi_j^2}{q_j^2} \equiv \phi_j^2$ . We then generate finite choice data  $(x_j^1, x_j^2)$  by simulating, for each menu  $A_j$ ,  $N$  independent choices (i) according to CRRA expected utility with  $t \sim \text{Log}(\tau = .6, \sigma = .3)$ , and (ii) according to CES utility with  $t \sim \text{Log}(\tau = .2, \sigma = .2)$ .<sup>42</sup> We study the effect of sample size by considering datasets with  $N = \{20, 80, 160, 480\}$  observations per menu.

Tables 1a and 1b report the ordered menus in our dataset, indexed by  $\phi_j = \phi_j^2/\phi_j^1$  and  $\pi_j = \pi_j^2/\pi_j^1$ , respectively. Figure 1 reports the empirical CDFs (ECDFs) of consumption ratios  $r_j(x) = x_j^2/x_j^1$  for risk, and  $v_j = 1 - e^{-r_j(x)}$  for altruism, in ten randomly selected menus. Note that the parametrization  $v_j$  in the altruism context is used for convenience such that  $v_j(x) \in [0, 1]$ .

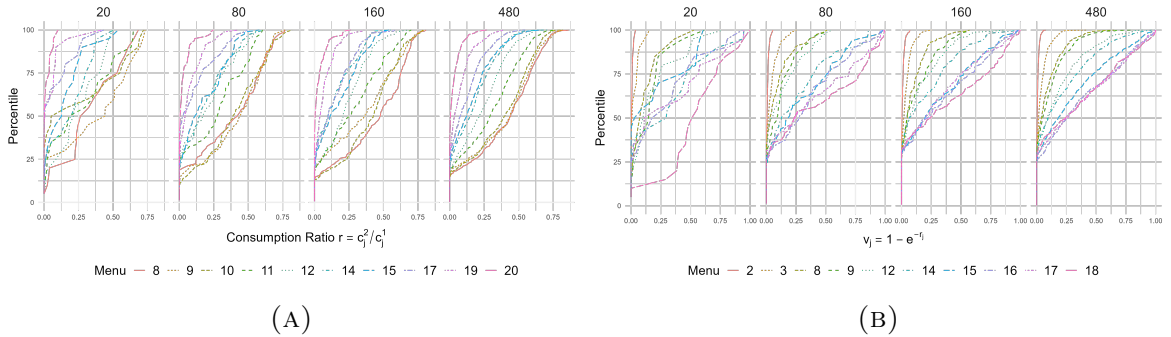
| Menu | $\phi_j$ | $\pi_j^1$ | $\pi_j^2$ | $q_j^1$ | $\phi_j^1$ | $\phi_j^2$ |
|------|----------|-----------|-----------|---------|------------|------------|
| 1    | 1.019    | .024      | .011      | .696    | .034       | .035       |
| 2    | 1.168    | .025      | .045      | .393    | .064       | .074       |
| 3    | 1.216    | .043      | .052      | .500    | .085       | .103       |
| 4    | 1.360    | .068      | .051      | .644    | .106       | .144       |
| 5    | 1.364    | .054      | .099      | .425    | .126       | .172       |
| 6    | 1.374    | .075      | .051      | .669    | .112       | .154       |
| 7    | 1.532    | .091      | .046      | .752    | .122       | .186       |
| 8    | 1.622    | .090      | .074      | .664    | .135       | .219       |
| 9    | 1.699    | .045      | .043      | .639    | .071       | .120       |
| 10   | 1.743    | .073      | .039      | .764    | .095       | .166       |

(A) Risk: Menus indexed by  $\phi_j$ 

| Menu | $\pi_j$ | $\pi_j^1$ | $\pi_j^2$ |
|------|---------|-----------|-----------|
| 1    | 7.064   | .010      | .071      |
| 2    | 6.208   | .013      | .081      |
| 3    | 2.512   | .026      | .066      |
| 4    | 1.977   | .046      | .091      |
| 5    | 1.963   | .045      | .088      |
| 6    | 1.856   | .039      | .073      |
| 7    | 1.805   | .025      | .045      |
| 8    | 1.331   | .051      | .068      |
| 9    | 1.234   | .078      | .097      |
| 10   | 1.056   | .056      | .059      |

(B) Altruism: Menus indexed by  $\pi_j$ 

TABLE 1. Summary of menu parameters under risk and altruism.

FIGURE 1. ECDFs of  $r_j$  in risk (left) and  $v_j = 1 - e^{-r_j}$  in altruism (right)

## D.2. Nonparametric Analysis, Theorem 1.

<sup>42</sup>These parameters give choice patterns (e.g., corner choices) that are in line with empirical observations in budget set experiments (see Choi, Kariv, Müller, and Silverman (2014) for the risk domain, and Andreoni and Miller (2002) for the altruism domain).

**Risk.** We first recall that the ORUM-rationalizability property in the risk domain is Quantile SAREU, the quantile version of the Strong Axiom of Revealed Expected Utility of Kubler, Selden, and Wei (2014). Quantile SAREU is satisfied if and only if for all quantiles  $c^p$  and all menu sequences  $\{j_1, \dots, j_K\}$  the loops  $L_p(j_1, j_2) \cdot L_p(j_2, j_3) \cdot \dots \cdot L_p(j_K, j_1) < 1$ . Verifying this property directly is computationally expensive, as the number of loops grows exponentially with  $J$ . To ease the verification process, we establish a connection between SAREU and the well-known problem of detecting negative cycles in a weighted directed graph. This allows us to leverage the existing literature on algorithms for identifying negative cycles.

Consider the complete directed graph where each node corresponds to one menu, and the weight of the directed edge from  $j$  to  $j'$  is

$$w(j, j') = \begin{cases} -\log L(j, j') & \text{whenever } L(j, j') > 0, \\ \infty & \text{otherwise.} \end{cases}$$

It is immediate that the logarithmic transformation and the change of sign guarantee that a violation of SAREU is equivalent to the existence of a negative cycle in the graph, i.e., a directed cycle where the sum of weights is negative. Now, in the context of a deterministic choice function  $c_j^p$ , notice that  $w(j, j')$  is determined exclusively by the ranking of consumption choices  $c_j^{p,1}, c_j^{p,2}, c_{j'}^{p,1}$  and  $c_{j'}^{p,2}$ , and that, given that  $\phi_j^1 < \phi_j^2$ , there are only six such possible rankings, as described in Table 2.

| Case                                                  | $w(j, j')$                                                                   | $w(j', j)$                                                                   |
|-------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| $c_j^{p,1} > c_j^{p,2} > c_{j'}^{p,1} > c_{j'}^{p,2}$ | $\log \phi_{j'}^1 - \log \phi_j^2 < 0$                                       | $\infty$                                                                     |
| $c_j^{p,1} > c_{j'}^{p,1} > c_j^{p,2} > c_{j'}^{p,2}$ | $\max\{\log \phi_{j'}^1 - \log \phi_j^1, \log \phi_{j'}^2 - \log \phi_j^2\}$ | $\log \phi_j^2 - \log \phi_{j'}^1 > 0$                                       |
| $c_j^{p,1} > c_{j'}^{p,1} > c_{j'}^{p,2} > c_j^{p,2}$ | $\log \phi_{j'}^1 - \log \phi_j^1$                                           | $\log \phi_j^2 - \log \phi_{j'}^2$                                           |
| $c_j^{p,1} > c_j^{p,1} > c_j^{p,2} > c_{j'}^{p,2}$    | $\log \phi_{j'}^2 - \log \phi_j^2$                                           | $\log \phi_j^1 - \log \phi_{j'}^1$                                           |
| $c_{j'}^{p,1} > c_j^{p,1} > c_{j'}^{p,2} > c_j^{p,2}$ | $\log \phi_{j'}^2 - \log \phi_j^1 > 0$                                       | $\max\{\log \phi_j^1 - \log \phi_{j'}^1, \log \phi_j^2 - \log \phi_{j'}^2\}$ |
| $c_{j'}^{p,1} > c_{j'}^{p,2} > c_j^{p,1} > c_j^{p,2}$ | $\infty$                                                                     | $\log \phi_j^1 - \log \phi_{j'}^2 < 0$                                       |

TABLE 2. Weights in the graph for menus  $A_j$  and  $A_{j'}$

Hence, given  $J$  menus, the directed graph associated with the deterministic case, with a single choice per menu, must be one of at most  $6 \frac{J^2}{2}$  types.<sup>43</sup> As an example, Figure 2 uses the weights in Table 2 to construct the directed graph for the median quantile. For presentation purposes, all edges in the graph have the same width, with black edges denoting negative weights. Black edges, those which can lead to violations of SAREU, are relatively scarce and decrease with the number of observations, as expected.

<sup>43</sup>This may be particularly useful from a computational point of view in cases where different sub-populations or datasets are available. Notice that it is feasible to determine which types of datasets are consistent with SAREU without the need for actual data and then verify whether actual data matches this structure.

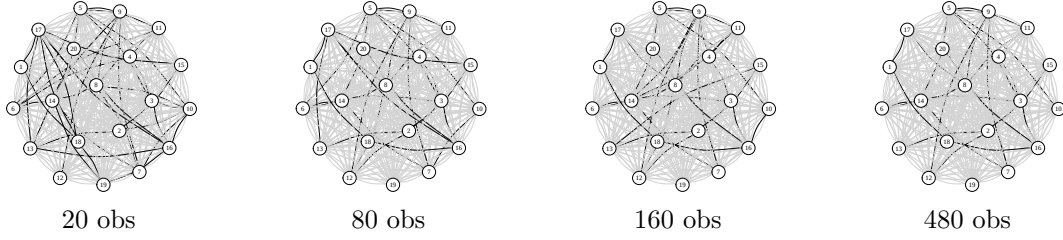


FIGURE 2. Directed graph using Table 2 for the median quantile and all menus

Once the graph has been constructed, we can search for negative cycles by applying well-known algorithms, such as the Bellman-Ford algorithm.<sup>44</sup>

We reproduce this exercise for the same 20 quantiles in all datasets, and Table 3 reports the non-rationalizable ones. The data have been generated by an ORUM; thus, as expected, one can immediately appreciate that the finer the sampling, the lower is the number of quantiles that are not rationalizable. Violations in small datasets are driven by the fact that the empirical quantiles  $\hat{c}^p$  differ from the true quantiles  $c^p$ . As  $N$  grows to infinity the number of violations will converge to 0.

| Dataset | Number of violations | Non-rationalizable quantiles                                       |
|---------|----------------------|--------------------------------------------------------------------|
| 20 obs  | 17                   | 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 95, 100 |
| 80 obs  | 11                   | 10, 15, 20, 25, 30, 35, 40, 45, 50, 95, 100                        |
| 160 obs | 8                    | 10, 15, 20, 25, 30, 35, 90, 95                                     |
| 480 obs | 6                    | 10, 15, 20, 25, 30, 35                                             |

TABLE 3. Non-rationalizable quantiles across datasets

**Altruism.** Just as ORUM-rationalizability reduces to Quantile SAREU in the risk application, it reduces to Quantile WARP in the altruism domain. The verification of WARP is simpler than that of SAREU, and so is its analysis. In a consumer setting, Kitamura and Stoye (2018) show that one can discretize choices to work with finite partitions, called patches, and Hoderlein and Stoye (2015) exploit this technique for the case of two goods. This approach, within our setting, reads as follows. Suppose that a quantile  $c^p$  fails to satisfy WARP. Then, there must exist two menus  $A_j, A_{j'}$ , with  $\pi_j > \pi_{j'}$ , where this violation occurs. Denoting by  $x_{j,j'}$  the allocation at which these two budget lines intersect, the violation guarantees that  $F_j(v_j(x_{j,j'})) < p \leq F_{j'}(v_{j'}(x_{j,j'}))$ . Then, the analysis of Quantile WARP simply requires to check  $F_j(v_j(x_{j,j'})) \geq F_{j'}(v_{j'}(x_{j,j'}))$  for every pair of menus. This verification only involves a few comparisons, so it can be performed directly. Table 4 reports the number of such WARP violations in our data.

Once again, we observe that the finer the sampling, the lower the number of violations. This is to be expected, as the data has indeed been generated under an ORUM, such that violations in small datasets are driven solely by  $\hat{F}_j \neq F_j$  and/or  $\hat{F}_{j'} \neq F_{j'}$ .

<sup>44</sup>See, e.g., Bang-Jensen and Gutin (2008) for a textbook treatment.

| $N$ obs        | Number of violations | Menus with positive share of violations |
|----------------|----------------------|-----------------------------------------|
| <b>20 obs</b>  | 3                    | 4, 13, 18                               |
| <b>80 obs</b>  | 2                    | 5, 8                                    |
| <b>160 obs</b> | 0                    | -                                       |
| <b>480 obs</b> | 0                    | -                                       |

TABLE 4. Non-rationalizable menus in altruism application

It is important to note that, although the heuristics adopted for analyzing both the risk and altruism cases are novel and foundational for the nonparametric econometric testing of ORUMs, a simple count of the number of violations does not constitute a valid statistical test.<sup>45</sup> In particular, the procedures described above do not account for sampling uncertainty. While Theorem 1 refers to the true data generating CDF  $F$ , we only have access to its empirical finite data counterpart  $\hat{F}$ , as a result, some of the violations could be due to sampling rather than failures of rationalizability. In Section D.5, we discuss in detail some of the challenges involved in designing asymptotically valid tests.

**D.3. Semi-parametric Analysis, Theorem 2.** Verifying the empirical restrictions of the type-utility maps under consideration, CRRA for risk and CES for altruism, is conveniently simple. In particular, these type-utility maps impose that the induced types  $t$  (risk aversion coefficients in the case of risk and altruism coefficients in the case of altruism) have to be constant across menu quantiles, or, equivalently, have the same empirical distribution. Whenever the utility maps contain any type independent meta-parameters, as is the case with the curvature  $\alpha$  in the altruism setting, the induced type distributions must be identical for the same meta-parameter values.

Under the CRRA and CES maps, the type-utility maps are invertible, thus one can recover directly the induced types from the choice data. The only difference in this stage is that, in the altruism setting, in order to recover  $t$  we first need to recover  $\alpha$ .  $\alpha$  is point identified by  $\alpha = \frac{\log(\pi_j/\pi_{j'})}{\log(r_j(c_j^p)/r_{j'}(c_{j'}^p))} + 1$  for all menus and all quantiles such that  $c_j^{p,2} > 0$ , so one can estimate  $\alpha$  as the median of these values in each dataset, and then recover  $t$  as normal. Table 5 presents the estimated values which, indeed, are very close to the true  $\alpha = .4$ .

|          | <b>20 obs</b> | <b>80 obs</b> | <b>160 obs</b> | <b>480 obs</b> | <b>True</b> |
|----------|---------------|---------------|----------------|----------------|-------------|
| $\alpha$ | .445          | .409          | .387           | .397           | .4          |

TABLE 5. Median  $\alpha$ 

<sup>45</sup>Nevertheless, a large number of non-rationalizable quantiles in reasonably sized datasets can still be suggestive of a failure of ORUM rationalizability.

Once the induced parameters  $t$  have been recovered, testing the empirical content of the CRRA and CES maps reduces to verifying that all menu samples belong to the same distribution. This logic can be implemented via a nonparametric equality of distributions test.<sup>46</sup> We do so through a  $J$ -sample adjusted version of the Anderson-Darling (AD) test. Intuitively, the AD test computes the quadratic distance between the empirical CDF and the hypothesized distribution imposing higher weights in the tails. The  $J$ -sample version of this test is a generalization of the standard 2 samples AD test to  $J$  samples (menus). Figure 3 plots the empirical CDFs of the induced risk aversion and altruism parameters, which can already be seen to be very similar. Table 6 verifies this intuition by reporting the results of the AD test. In this setting the null is that all samples belong to the same distribution, thus a small p-value ( $\leq .05$ ) would suggest that not all samples belong to the same distribution. The exceptionally high p-values reported in Table 6 suggest that it is very likely that all samples indeed come from the same distribution, i.e., CRRA and CES prescriptions seem to be satisfied in the data.<sup>47</sup>

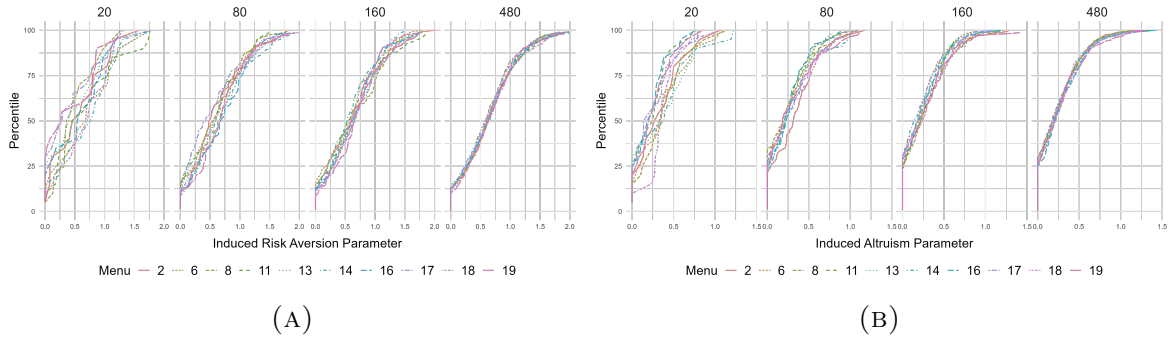


FIGURE 3. ECDF of induced risk aversion (left) and altruism (right) parameters

|         | Risk         |         | Altruism     |         |
|---------|--------------|---------|--------------|---------|
| N obs   | AD Statistic | p-value | AD Statistic | p-value |
| 20 obs  | 17.499       | 0.655   | 17.189       | 0.693   |
| 80 obs  | 5.417        | 1.000   | 4.222        | 1.000   |
| 160 obs | 3.364        | 1.000   | 2.906        | 1.000   |
| 480 obs | 1.586        | 1.000   | 1.617        | 1.000   |

TABLE 6.  $J$ -sample adjusted Anderson-Darling test results for risk and altruism

<sup>46</sup>There are several tests of this kind available in the literature with different power and properties. Some of them include the Kolmogorov-Smirnov test, the Anderson-Darling test, the Cramer-von Mises test or the Dudley-Tucker-Smith test. We select the  $J$ -sample adjusted version of the Anderson-Darling test due to its popularity, high power, robustness and good tail behavior. See D’Agostino and Stephens (1986) for a classic reference or Hollander, Wolfe and Chicken (2013) for a more recent treatment.

<sup>47</sup>We should note that the AD test is not accounting for the fact that the meta-parameter  $\alpha$  has been estimated. We discuss this issue further in Section D.5.

**D.4. Parametric Analysis, Theorem 3.** Recall that the parametric testing in our application reduces to verifying if the induced parameters  $t$  have been generated under a (left censored at  $t < 0$ ) logistic distribution. We discuss two complementary approaches on this regard. As we saw in Corollaries 3 and 5,  $F$  belongs to the logistic family of distributions  $\mathcal{G}^L$  if and only if it satisfies the context-dependent CE and CLA properties. The analyst may verify these properties directly in what we call the axiom-based approach. In addition, one may move away from the axiomatic characterization, and adopt a statistical approach to the problem. One such alternative is to first estimate the location and scale parameters of a censored logistic via maximum-likelihood and then run a nonparametric goodness-of-fit test via parametric bootstrap. By doing so, the researcher is asking how close the data are to the best-fitting logistic distribution. The first approach is closer to the spirit and results of the paper, while the second is more grounded in state-of-the-art econometrics and does not hinge on the if and only if conditions. As a result, we see them as complementary.

**Axiom-based approach.** CE and CLA impose very clear and testable properties. CE requires some mass at  $r = 0$  ( $v = 0$ ) and no mass at  $r = 1$  ( $v = 1$ ) in the context of risk (altruism). Tables 7 and 9 report the share (%) of observations with  $r$  ( $v$ ) larger than 0 for each menu and dataset. They are all strictly positive as prescribed by CE, and as to be expected if data had been generated under a distribution with continuous support left-truncated at 0 (like censored logit). Truncated observations would be those with parameter  $t$  less or equal 0.

Tables 8 and 10 report the largest value of  $r$  (and  $v$ ) of each menu, which can all be seen as being strictly less than 1. Most of these values are relatively large, with some exceptions that, intuitively, correspond to menus with very high relative price of the second state.

| Menu | $\phi_j$ | 20 obs | 80 obs | 160 obs | 480 obs |
|------|----------|--------|--------|---------|---------|
| 1    | 1.019    | 5      | 8.75   | 9.375   | 13.958  |
| 2    | 1.168    | 5      | 10.00  | 15.000  | 12.292  |
| 3    | 1.216    | 15     | 13.75  | 17.500  | 11.875  |
| 4    | 1.360    | 10     | 16.25  | 12.500  | 10.625  |
| 5    | 1.364    | 15     | 21.25  | 8.750   | 10.833  |
| 6    | 1.374    | 10     | 13.75  | 15.625  | 11.667  |
| 7    | 1.532    | 10     | 10.00  | 10.000  | 12.500  |
| 8    | 1.622    | 5      | 15.00  | 11.250  | 12.500  |
| 9    | 1.699    | 20     | 11.25  | 14.375  | 13.750  |
| 10   | 1.743    | 25     | 6.25   | 9.375   | 11.875  |
| 11   | 2.151    | 10     | 15.00  | 13.125  | 13.125  |
| 12   | 2.672    | 0      | 16.25  | 9.375   | 12.292  |
| 13   | 3.241    | 25     | 11.25  | 10.000  | 13.125  |
| 14   | 3.445    | 15     | 8.75   | 12.500  | 11.458  |
| 15   | 3.457    | 5      | 12.50  | 10.625  | 11.042  |
| 16   | 3.832    | 20     | 10.00  | 13.125  | 12.500  |
| 17   | 4.451    | 30     | 16.25  | 11.250  | 12.500  |
| 18   | 8.178    | 10     | 12.50  | 11.250  | 12.500  |
| 19   | 8.343    | 30     | 10.00  | 8.750   | 9.792   |
| 20   | 21.841   | 15     | 8.75   | 15.000  | 11.042  |

TABLE 7. Risk: Share of observations with  $r = 0$ 

| Menu | $\phi_j$ | 20 obs | 80 obs | 160 obs | 480 obs |
|------|----------|--------|--------|---------|---------|
| 1    | 1.019    | .989   | .990   | .990    | .992    |
| 2    | 1.168    | .906   | .917   | .925    | .946    |
| 3    | 1.216    | .905   | .895   | .924    | .925    |
| 4    | 1.360    | .777   | .857   | .883    | .888    |
| 5    | 1.364    | .835   | .880   | .894    | .880    |
| 6    | 1.374    | .836   | .879   | .837    | .891    |
| 7    | 1.532    | .784   | .842   | .843    | .832    |
| 8    | 1.622    | .684   | .767   | .810    | .863    |
| 9    | 1.699    | .733   | .765   | .783    | .810    |
| 10   | 1.743    | .721   | .818   | .802    | .822    |
| 11   | 2.151    | .648   | .604   | .702    | .767    |
| 12   | 2.672    | .496   | .577   | .684    | .652    |
| 13   | 3.241    | .463   | .531   | .615    | .632    |
| 14   | 3.445    | .505   | .610   | .567    | .713    |
| 15   | 3.457    | .537   | .582   | .541    | .651    |
| 16   | 3.832    | .343   | .535   | .561    | .592    |
| 17   | 4.451    | .433   | .494   | .581    | .547    |
| 18   | 8.178    | .203   | .379   | .406    | .488    |
| 19   | 8.343    | .418   | .431   | .361    | .387    |
| 20   | 21.841   | .102   | .248   | .264    | .270    |

TABLE 8. Risk: Largest  $r$  across menus

| Menu | $\pi_j$ | 20 obs | 80 obs | 160 obs | 480 obs |
|------|---------|--------|--------|---------|---------|
| 1    | 7.064   | 25     | 26.25  | 23.750  | 24.375  |
| 2    | 6.208   | 20     | 21.25  | 25.000  | 26.042  |
| 3    | 2.512   | 35     | 21.25  | 24.375  | 25.417  |
| 4    | 1.977   | 25     | 28.75  | 27.500  | 23.750  |
| 5    | 1.963   | 30     | 20.00  | 22.500  | 28.542  |
| 6    | 1.856   | 15     | 28.75  | 28.125  | 30.000  |
| 7    | 1.805   | 25     | 21.25  | 24.375  | 27.917  |
| 8    | 1.331   | 15     | 33.75  | 23.750  | 29.167  |
| 9    | 1.234   | 10     | 21.25  | 28.125  | 26.875  |
| 10   | 1.056   | 35     | 23.75  | 22.500  | 26.458  |
| 11   | .958    | 25     | 27.50  | 23.750  | 26.250  |
| 12   | .824    | 15     | 28.75  | 25.000  | 24.583  |
| 13   | .821    | 30     | 26.25  | 26.875  | 28.125  |
| 14   | .679    | 25     | 22.50  | 30.625  | 29.792  |
| 15   | .541    | 35     | 25.00  | 28.125  | 23.958  |
| 16   | .444    | 25     | 22.50  | 28.750  | 25.000  |
| 17   | .444    | 25     | 31.25  | 31.250  | 26.042  |
| 18   | .402    | 10     | 25.00  | 30.000  | 28.125  |
| 19   | .353    | 20     | 23.75  | 30.625  | 26.458  |
| 20   | .231    | 25     | 21.25  | 26.875  | 26.875  |

| Menu | $\phi_j$ | 20 obs | 80 obs | 160 obs | 480 obs |
|------|----------|--------|--------|---------|---------|
| 1    | 7.064    | .045   | .045   | .045    | .058    |
| 2    | 6.208    | .037   | .056   | .072    | .077    |
| 3    | 2.512    | .154   | .254   | .255    | .421    |
| 4    | 1.977    | .294   | .394   | .415    | .499    |
| 5    | 1.963    | .168   | .359   | .310    | .566    |
| 6    | 1.856    | .310   | .294   | .330    | .585    |
| 7    | 1.805    | .213   | .464   | .613    | .629    |
| 8    | 1.331    | .528   | .510   | .574    | .649    |
| 9    | 1.234    | .616   | .499   | .570    | .794    |
| 10   | 1.056    | .617   | .648   | .860    | .804    |
| 11   | .958     | .486   | .701   | .737    | .888    |
| 12   | .824     | .772   | .555   | .862    | .889    |
| 13   | .821     | .640   | .971   | .796    | .894    |
| 14   | .679     | .949   | .859   | .993    | .965    |
| 15   | .541     | .607   | .978   | .985    | .995    |
| 16   | .444     | .935   | .969   | .994    | .999    |
| 17   | .444     | .947   | .990   | .989    | 1.000   |
| 18   | .402     | .998   | .995   | 1.000   | 1.000   |
| 19   | .353     | .994   | 1.000  | 1.000   | 1.000   |
| 20   | .231     | .996   | 1.000  | 1.000   | 1.000   |

TABLE 9. Altruism: Share of observations with  $v_j = 0$       TABLE 10. Altruism: Largest  $v_j$  across menus

To test for CLA, we consider all triples formed by one menu  $A_j$  and two interior observations in this menu.<sup>48</sup> Take the induced risk aversion (altruism) coefficients corresponding to these two observations and compute their sum. Take the empirical quantiles of these two observations, transform them into log-odds and compute its sum. CLA requires the relationship between these two sums to be a linear function. Figure 4 plots this relationship for all pairs formed by combining 5 randomly selected interior choices per menu, and shows that this relationship is clearly linear.

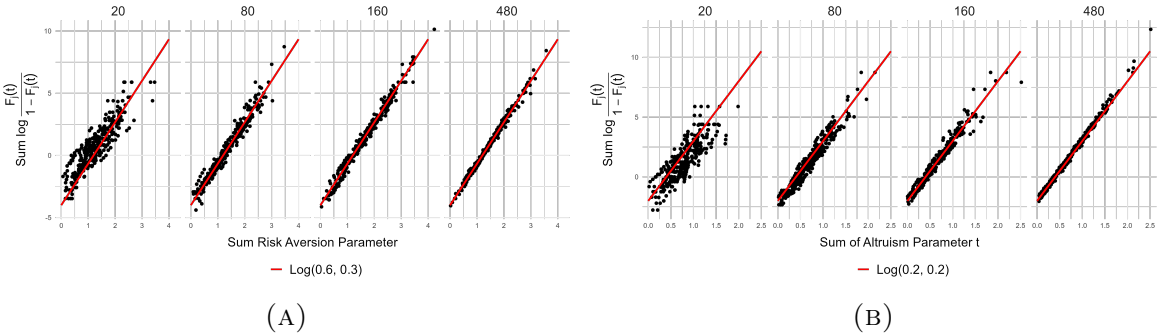


FIGURE 4. CLA property in risk application (left) and altruism (right)

To conclude the theory approach, we emphasize that this method is not testing CE and CLA in a statistical sense. Instead, it is simply showing intuitive data evidence consistent with the theoretical prescriptions. We now move away from the theory and adopt a statistical approach to the problem.

<sup>48</sup>Given the finiteness, we also need to exclude the largest observation in each menu.

**Statistical Approach.** There exist several (non-parametric) methods to test model misspecification, i.e., if  $F$  belongs to the logistic family. We follow a standard, popular and powerful test, a parametric-bootstrap adaptation of the Anderson-Darling test (AD). However, we acknowledge that this procedure is not unique. Moreover, it is not directly related to the theoretical results in the main text, and, as such, we only describe its implementation at a high-level. Intuitively, we compare  $F$  to the distribution in the logistic family which is closest to  $F$  in a maximum-likelihood sense. In particular, we first use the induced  $t$  values to estimate the location and scale parameters of the logistic via maximum-likelihood, i.e.,  $(\hat{\tau}^{\text{MLE}}, \hat{\sigma}^{\text{MLE}})$ ; and then we compute an AD test statistic ( $\text{AD}^{\text{MLE}}$ ) against the best-fitting logistic  $\text{Log}(\hat{\tau}^{\text{MLE}}, \hat{\sigma}^{\text{MLE}})$  at the induced values.

The test above does not account for the fact that  $\hat{\tau}^{\text{MLE}}$  and  $\hat{\sigma}^{\text{MLE}}$  have been estimated, i.e.,  $(\hat{\tau}^{\text{MLE}}, \hat{\sigma}^{\text{MLE}}) \neq (\tau, \sigma)$ , so standard inference is not immediately available. To obtain asymptotically correct p-values, we adopt a standard parametric bootstrap approach. More precisely, we draw  $B = 1000$  samples of  $N$  observations of  $t$  from the null-model  $\text{Log}(\hat{\tau}^{\text{MLE}}, \hat{\sigma}^{\text{MLE}})$ , we re-estimate the logistic parameters via MLE using the bootstrap sample  $(\hat{\tau}_b^{\text{MLE}}, \hat{\sigma}_b^{\text{MLE}})$ , and, finally, we compute the associated AD statistic ( $\text{AD}_b$ ) between  $\text{Log}(\hat{\tau}^{\text{MLE}}, \hat{\sigma}^{\text{MLE}})$  and  $\text{Log}(\hat{\tau}_b^{\text{MLE}}, \hat{\sigma}_b^{\text{MLE}})$ . We set the p-value equal to the fraction of  $\text{AD}_b$  test statistics which are higher than the initial  $\text{AD}^{\text{MLE}}$ , i.e.,  $\frac{1}{B} \sum_{b=1}^B 1(\text{AD}_b \geq \text{AD}^{\text{MLE}})$ . By re-estimating  $\tau$  and  $\sigma$  in each bootstrap draw, we account for the uncertainty coming from parameter estimation. A p-value below .05, as it is customary, would suggest that  $F$  does not belong to the logistic family.

The log-likelihood has a familiar closed form. Let  $1 \leq m \leq M$  be a generic observation with  $M = J \times N$ . Let  $m \in \{1, \dots, M_0\}$  be the set of observations with consumption ratio equal to 0. All such  $m$  must have  $t \leq 0$ , i.e., a negative relative risk aversion coefficient or a positive altruism coefficient. For convenience, define the normalized values  $\bar{0} = \frac{0-\tau}{\sigma}$  and  $\bar{t}_m = \frac{t_m-\tau}{\sigma}$ , where  $\tau$  is the location parameter and  $\sigma$  the scale parameter of the logistic. The log-likelihood can be written as:<sup>49</sup>

$$\begin{aligned} & \sum_1^{M_0} \log G^{(\tau, \sigma)}(0) + \sum_{M_0+1}^M \log g^{(\tau, \sigma)}(t_m) = \\ & -M_0 \log(1 + e^{-\bar{0}}) - \sum_{m=M_0+1}^M \left( \bar{t}_m + \log \sigma + 2 \log(1 + e^{-\bar{t}_m}) \right) \end{aligned}$$

Table 11 reports the MLE estimators and the results of the AD test, conditional on the number of observations, for risk and altruism. As we can see, the MLE fits the values very closely to the true values and the AD test reports very high p-values. We cannot reject the null in any of the specifications, which suggests that values could have been generated from  $\text{Log}(\hat{\tau}^{\text{MLE}}, \hat{\sigma}^{\text{MLE}})$ , and, as a result, that  $F$  belongs to the

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<sup>49</sup> $g^{(\tau, \sigma)}$  denotes the density of  $G^{(\tau, \sigma)}$ .

logistic family. We note that the fit of the altruism values is slightly worse than their risk analogues. This is to be expected as the altruism application involved the pre-estimation of the meta-parameter  $\alpha$  as discussed in the semi-parametric section.

|         | Risk         |                | Altruism     |                |
|---------|--------------|----------------|--------------|----------------|
| N obs   | $\hat{\tau}$ | $\hat{\sigma}$ | $\hat{\tau}$ | $\hat{\sigma}$ |
| 20 obs  | .567         | .323           | .264         | .207           |
| 80 obs  | .605         | .315           | .225         | .199           |
| 160 obs | .606         | .308           | .200         | .196           |
| 480 obs | .606         | .305           | .199         | .197           |
| True    | .6           | .3             | .2           | .2             |

|         | Risk    |          | Altruism |          |
|---------|---------|----------|----------|----------|
| N obs   | AD Stat | p-values | AD Stat  | p-values |
| 20 obs  | 11.280  | .716     | 26.406   | .171     |
| 80 obs  | 37.997  | .736     | 114.654  | .234     |
| 160 obs | 72.357  | .750     | 248.886  | .401     |
| 480 obs | 225.010 | .423     | 751.177  | .433     |

TABLE 11. Parameter estimation via Maximum Likelihood (left) and AD Test via Parametric Bootstrap (right)

**D.5. Limitations and Next Steps.** The particularization of the theory to concrete economic applications, the computational exercise involved in the non-parametric exercise within the risk application, and the adoption of standard tools for the development of efficient tests in the semi and parametric contexts are all contributions that may be leveraged in future econometric work. At the same time, we have emphasized throughout some of the limitations of our empirical illustration. Although addressing these challenges lies beyond the scope of this paper, which has focused on the choice-based foundations of ORUMs under various parametric assumptions, we now provide some initial ideas in this regard.

An important direction for future work is the development of asymptotically valid and efficient tests in the nonparametric framework. Our current approach does not account for sampling uncertainty, which can lead to violations even when the data are generated from an ORUM. We conjecture that in more amenable applications, such as the altruism one, the abundant theory on multiple testing of moment inequalities could be leveraged to construct valid tests.<sup>50</sup> However, in other applications, like the case of risk, the specific rationalizability conditions are more involved, making unlikely the

<sup>50</sup>For instance, note that in the altruism domain, testing Quantile WARP reduces to verify  $F_j(v_j(x_{j,j'})) \leq F_{j'}(v_{j'}(x_{j,j'}))$  for every pair of menus. One could then consider  $\hat{F}_j(v_j(x_{j,j'}))$ ,  $\hat{F}_{j'}(v_{j'}(x_{j,j'}))$  as random variables, and re-write the testing hypothesis as  $E[\hat{F}_j(v_j(x_{j,j'})) - \hat{F}_{j'}(v_{j'}(x_{j,j'}))] \geq 0$ . These moment inequalities can then be stacked to obtain a system of moment inequalities. To overcome the obvious limitation of having just one observation  $\hat{F}_j(v_j(x_{j,j'}))$ , what will preclude any form of statistical testing, we may draw  $B$  bootstrap sub-samples with repetition to create a sequence of *iid* random variables  $\hat{F}_j^b(v_j(x_{j,j'}))$ . There exists a promising avenue for devising tests in this context leveraging the ideas in Chernozhukov, Chetverikov and Kato (2016), although their methods are not immediately applicable to our context. A simple and asymptotically valid alternative, yet not very powerful, would be to conduct a permutation based test under the strong null  $F_j = F_{j'}$ . In our simulated dataset, this test leads to qualitatively similar results to the ones presented in the relevant section. We can provide more details upon request.

derivation of a single efficient test across all settings. In addition, computational constraints also suggest the need for efficient alternatives to standard resampling methods.

The landscape is more positive in the semi- and parametric settings, at least for the applications under consideration. In this case, standard tools are immediately available with the only challenge of incorporating the uncertainty from estimating meta-parameters like  $\alpha$  in the altruism case. The required adjustments seem to be standard, though they may be algebraically involved and context specific. To summarize, in the prevalent applications to risk and altruism, our simulation exercise provides valid semi and parametric tests for the ORUM rationalizability framework, albeit minor adjustments. Moreover, it lays down a promising terrain for the development of statistically valid tests in the non-parametric setting.

Finally, exploring richer type-utility maps is a promising direction. This may involve working with implicit types, estimating additional parameters, or adopting alternative testing strategies beyond distributional comparisons.

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